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MTH 111 — Past Questions & Materials
by Renance

Course: MTH 111
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 100

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Paper 1 of 3 — MTH 111 answers 2017/2018

Level 100 • First Semester • PastQuestion

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MTH 111 EXAM QUESTIONS ANSWER — 2017/18 SESSION

$$\frac{1 + \tan 60 \tan 30}{\tan 60 \tan 30}$$

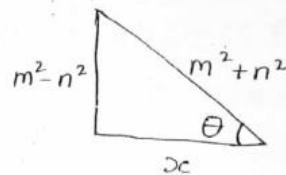
Recall $\tan 60 = \sqrt{3}$; $\tan 30 = \frac{1}{\sqrt{3}}$

$$\therefore \frac{1 + \tan 60 \tan 30}{\tan 60 \tan 30} = \frac{1 + \sqrt{3} \cdot \frac{1}{\sqrt{3}}}{\sqrt{3} \cdot \frac{1}{\sqrt{3}}} = \frac{1+1}{1}$$

2 No Correct Option.

Given $\sin \theta = \frac{m^2 - n^2}{m^2 + n^2}$

$$\therefore \cot \theta = \frac{1}{\tan \theta}$$



$$\begin{aligned} x^2 &= (m^2 + n^2)^2 - (m^2 - n^2)^2 \\ &= m^4 + 2m^2n^2 + n^4 - [m^4 - 2m^2n^2 + n^4] \\ &= m^4 + 2m^2n^2 + n^4 - m^4 + 2m^2n^2 - n^4 \end{aligned}$$

$$x^2 = 4m^2n^2$$

$$x = \sqrt{4m^2n^2} = 2mn$$

$$\therefore \cot \theta = \frac{1}{\tan \theta} = \frac{1}{\frac{m^2 - n^2}{2mn}} = \frac{2mn}{m^2 - n^2}$$

A

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$$\frac{1 + \tan 60}{1 - \tan 60} = \frac{1 + \sqrt{3}}{1 - \sqrt{3}}$$

Recall $\tan 60 = \sqrt{3}$

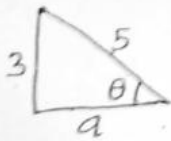
$$\frac{1 + \sqrt{3}}{1 - \sqrt{3}} \times \frac{1 + \sqrt{3}}{1 + \sqrt{3}} \quad \left[\text{Multiply both num and den with the conjugate of the denominator} \right]$$

$$= \frac{1 + 2\sqrt{3} + 3}{1^2 - (\sqrt{3})^2} = \frac{4 + 2\sqrt{3}}{1 - 3} = \frac{4 + 2\sqrt{3}}{-2}$$

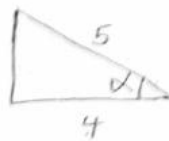
$$= -2 - \sqrt{3}$$

Closest Option **D**

Given $\sin \theta = \frac{3}{5}$; $\sin \alpha = \frac{3}{5}$; $\cos \theta = \frac{4}{5} \Rightarrow \frac{4}{5}$
 $\cos \alpha = \frac{4}{5}$



$$\begin{aligned} a^2 &= 5^2 - 3^2 \\ &= 25 - 9 \\ &= 16 \\ a &= \sqrt{16} = 4 \end{aligned}$$



$$\begin{aligned} b^2 &= 5^2 - 4^2 \\ &= 25 - 16 \\ &= 9 \\ b &= \sqrt{9} = 3 \end{aligned}$$

B

$$\begin{aligned} \therefore \sin(\theta + \alpha) &= \sin \theta \cos \alpha + \cos \theta \sin \alpha \\ &= \left(\frac{3}{5} \times \frac{4}{5} \right) + \left(\frac{4}{5} \times \frac{3}{5} \right) \\ &= \frac{12}{25} + \frac{12}{25} = \frac{12+12}{25} = \frac{24}{25} \end{aligned}$$

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Given $\cos x = \frac{\sqrt{a}}{\sqrt{b}}$

$\csc x = \frac{1}{\sin x}$

$\sin x = \frac{\text{opp}}{\text{hyp}} = \frac{d}{\sqrt{b}} = \frac{\sqrt{b-a}}{\sqrt{b}}$ (A)

$\therefore \csc x = \frac{1}{\frac{\sqrt{b-a}}{\sqrt{b}}} = \frac{\sqrt{b}}{\sqrt{b-a}} = \sqrt{\frac{b}{b-a}}$

Given $\sin(x+y) = \frac{\sqrt{2}}{2}$ --- (1)

$\cos 2x = -\frac{1}{2}$ --- (2)

Using eqn (2)

$2x = \cos^{-1}(-\frac{1}{2}) = 120$

$x = \frac{120}{2} = 60^\circ$ (C)

Using eqn (1)

$\sin(x+y) = \frac{\sqrt{2}}{2}$

$x+y = \sin^{-1}(\frac{\sqrt{2}}{2}) = 45$ --- (3)

Put $x = 60^\circ$ into eqn (3)

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$$60^\circ + y = 45^\circ$$

$$y = 45^\circ - 60^\circ = -15^\circ$$

$$\therefore x = 60^\circ, y = -15^\circ$$

$$7 \quad \sec \theta \cot \theta = \frac{1}{\cos \theta} \times \frac{\cos \theta}{\sin \theta} = \frac{1}{\frac{1}{\cos \theta} \times \frac{1}{\tan \theta}}$$

$$\text{Recall that; } \tan \theta = \frac{2 \tan \frac{1}{2} \theta}{1 - \tan^2 \frac{1}{2} \theta}$$

$$\begin{aligned} \cos \theta &= \cos \left(\frac{1}{2} \theta + \frac{1}{2} \theta \right) \\ &= \frac{\cos^2 \left(\frac{1}{2} \theta \right) - \sin^2 \left(\frac{1}{2} \theta \right)}{1} \end{aligned}$$

$$= \frac{\cos^2 \frac{1}{2} \theta - \sin^2 \frac{1}{2} \theta}{\cos^2 \frac{1}{2} \theta + \sin^2 \frac{1}{2} \theta}$$

Divide both num & Den. through by $\cos^2 \frac{1}{2} \theta$

$$= \frac{1 - \tan^2 \frac{1}{2} \theta}{1 + \tan^2 \frac{1}{2} \theta}$$

$$\text{Finally, } \frac{1}{\cos \theta} \times \frac{1}{\tan \theta} = \frac{1 + \tan^2 \frac{1}{2} \theta}{1 - \tan^2 \frac{1}{2} \theta} \times \frac{1 - \tan^2 \frac{1}{2} \theta}{2 \tan \frac{1}{2} \theta}$$

$$\text{where } \tan \frac{1}{2} \theta = t$$

$$\therefore \frac{1+t^2}{1-t^2} \times \frac{1-t^2}{2t} = 1$$



$$8 \quad \frac{1+t^2}{\cancel{1-t^2}} \times \frac{\cancel{1-t^2}}{2t} = 1$$

$$\frac{1+t^2}{2t} = 1 \Rightarrow 2t = 1+t^2$$

$$t^2 - 2t + 1 = 0$$

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By solving quadratically

$$t^2 - t - t + 1 = 0$$

$$t(t-1) - 1(t-1) = 0$$

$$(t-1)(t-1) = 0$$

$$\therefore t-1 = 0 \text{ or } t-1 = 0$$

$$\therefore t = 1 \text{ twice}$$

(B)

$$\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = 1$$

$$\frac{2t + 1 - t^2}{1 + t^2} = 1$$

$$2t + 1 - t^2 = 1 + t^2$$

$$1 + t^2 + t^2 - 1 - 2t = 0$$

$$2t^2 - 2t = 0$$

$$2t(t-1) = 0$$

$$\text{Either } 2t = 0 \text{ or } t-1 = 0$$

$$t = \frac{0}{2} = 0$$

$$t = 1$$

$$\therefore t = 0 \text{ or } 1$$

(A)

$$3 \tan \theta = 2 \cos \theta$$

$$\frac{\tan \theta}{\cos \theta} = \frac{2}{3}$$

$$\frac{\sin \theta}{\cos \theta} \cdot \frac{1}{\cos \theta} = \frac{2}{3}$$

$$\left[\text{Note: } \tan \theta = \frac{\sin \theta}{\cos \theta} \right]$$

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$$\frac{\sin \theta}{\cos^2 \theta} = \frac{2}{3}$$

Recall that: $\cos^2 \theta = 1 - \sin^2 \theta$

$$\frac{\sin \theta}{1 - \sin^2 \theta} = \frac{2}{3}$$

$$2 - 2\sin^2 \theta = 3\sin \theta$$

$$2\sin^2 \theta + 3\sin \theta - 2 = 0$$

$$\text{Let } \sin \theta = b$$

$$2b^2 + 3b - 2 = 0$$

$$2b^2 + 4b - b - 2 = 0$$

$$2b(b+2) - 1(b+2) = 0$$

$$(2b-1)(b+2) = 0$$

$$\text{Either } 2b-1=0 \text{ or } b+2=0$$

$$2b=1 \text{ or } b=-2$$

$$b = \frac{1}{2}$$

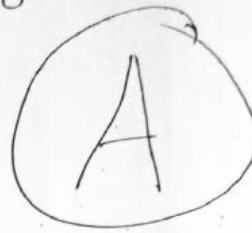
$$\text{Taking } b = \frac{1}{2} \Rightarrow \sin \theta = \frac{1}{2}$$

$$\theta = \sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$$

Sine is still positive in the 2nd Quadrant

$$\therefore \theta = 180^\circ - 30^\circ = 150^\circ$$

$$\Rightarrow \theta = 30^\circ \text{ or } \underline{\underline{150^\circ}}$$



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$$(2+3x)^5 = 1(2)^5 + 5(2)^4(3x) + 10(2)^3(3x)^2 + 10(2)^2(3x)^3 + 5(2)(3x)^4 + 1(3x)^5$$

$$= 32 + 240x + 720x^2 + 1080x^3 + 810x^4 + 243x^5$$

∴ Since Kx^4 is a term in the above equation

$$K = 810 //$$

(A)

$$\frac{\sqrt{6} - \sqrt{4}}{\sqrt{6} + \sqrt{4}}$$

$$\frac{\sqrt{6} - \sqrt{4}}{\sqrt{6} + \sqrt{4}} \times \frac{\sqrt{6} - \sqrt{4}}{\sqrt{6} - \sqrt{4}} = \frac{6 - \sqrt{24} - \sqrt{24} + 4}{2} = \frac{10 - 2\sqrt{24}}{2} =$$

$$= \frac{10 - 2 \times 2\sqrt{6}}{2} = \frac{10 - 4\sqrt{6}}{2} = 5 - 2\sqrt{6} //$$

(D)

B.

from $\sum_{n=1}^n (2n-1)^2$ the terms = $1^2 + 3^2 + 5^2 + 7^2 + 9^2 + \dots$
at $n=1 \quad 2 \quad 3 \quad 4 \quad 5$

By Comparing - the sum that fits into the terms

$$is = \left(\frac{n(n+1)}{2} \right)^2$$

B

$$(a-2b)^8$$

from Pascal's Triangle

$$8 \quad 28 \quad 56 \quad 70 \quad 56 \quad 28 \quad 8 \quad 1$$

↑
This is the middle term

expanding the equation

$$8 \quad -8 \quad -7 \quad -1 \quad -6 \quad -2 \quad 5 \quad 3 \quad 1 \quad 16$$

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10 $a^7(16)b^4 = +1120a^7b^4$ (B)

15 $(1-2x)^{-2}$

$$(1+x)^n = 1 + nx + \frac{n(n-1)x^2}{2!} + \dots$$

Given:
 $x = -2x$
 $n = -2$

3rd term

$$= 1 + (-2)(-2x) + \frac{(-2)(-2-1)(-2x)^2}{2 \times 1}$$

$$= 1 + 4x + \frac{(-2)(-3)(4x^2)}{2} = 1 + 4x + \frac{24x^2}{2} = 1 + 4x + 12x^2$$

$12x^2$ is the 3rd term (A)

6 The series is an Arithmetic progression (A.P.)
 The sum of an A.P is $S_n = \frac{n}{2} (2a + (n-1)d)$

Given that: from the A.P, the progression is $1^2 + 2^2 + 3^2 \dots$

$a = 1$
 $d = T_2 - T_1 = 2 - 1 = 1$

$$S_n = \frac{n}{2} (2a + (n-1)d) = \frac{n}{2} [2 + n - 1] = \frac{n}{2} [n + 1]$$
 (C)

2 Given $(1+2x)^m$ in the ratio $x^4 : x^3$

$$T_{r+1} = {}^m C_r \cdot a^{n-r} \cdot b^r \text{ where } a = 1, b = 2x$$

$${}^m C_r \cdot 1^{m-r} (2x)^r = {}^m C_r \cdot 2^r \cdot x^r$$

By Comparing Coefficients.

$$x^r = x^4 \quad ; \quad x^r = x^3$$

$$\Rightarrow r = 4 \quad ; \quad r = 3$$

(D)

$$\therefore \frac{{}^m C_4 \cdot 2^4 \cdot x^4}{{}^m C_3 \cdot 2^3 \cdot x^3} = \frac{3}{1} \Rightarrow \frac{{}^m C_4 \cdot 16}{{}^m C_3 \cdot 8} = \frac{3}{1}$$

$${}^m C_r = \frac{m!}{(m-r)!r!} = \frac{16m!}{(m-4)!4 \times 3!} = 24 \cdot \frac{m!}{(m-3)(m-4)!3!}$$

$$4 = \frac{24}{m-3} \Rightarrow m-3 = 6$$

$$m = 6 + 3 = 9$$

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$$\begin{aligned}
 & (3 + \sqrt{5})^3 - (3 - \sqrt{5})^3 \\
 &= (3 + \sqrt{5})^2(3 + \sqrt{5}) - [(3 - \sqrt{5})^2(3 - \sqrt{5})] \\
 &= (9 + 6\sqrt{5} + 5)(3 + \sqrt{5}) - [(9 - 6\sqrt{5} + 5)(3 - \sqrt{5})] \\
 &= (14 + 6\sqrt{5})(3 + \sqrt{5}) - [(14 - 6\sqrt{5})(3 - \sqrt{5})] \\
 &= 42 + 14\sqrt{5} + 18\sqrt{5} + 30 - [42 - 14\sqrt{5} - 18\sqrt{5} + 30] \\
 &= 42 + 32\sqrt{5} + 30 - 42 + 32\sqrt{5} - 30 \\
 &= 32\sqrt{5} + 32\sqrt{5} = (32 + 32)\sqrt{5} = \underline{64\sqrt{5}} \quad \text{(C)}
 \end{aligned}$$

n -positive integers are like 1, 2, 3, 4, 5, ...
 $a = 1, d = 1$
 $S_n = \frac{n}{2}(2a + (n-1)d) \Rightarrow \frac{n}{2}(2 + (n-1))$
 $= \frac{n}{2}(n+1) \quad \text{(C)}$

$$\begin{aligned}
 5\sqrt{3} - \sqrt{47} + 2\sqrt{m} &= 0 \\
 2\sqrt{m} &= \sqrt{49 \times 3} - 5\sqrt{3} \\
 2\sqrt{m} &= 7\sqrt{3} - 5\sqrt{3} \\
 &= (7-5)\sqrt{3} \\
 2\sqrt{m} &= 2\sqrt{3} \quad \text{(B)} \\
 \Rightarrow m &= \underline{3}
 \end{aligned}$$

$$\begin{aligned}
 x^2 + 4x + 40 &= 0 \\
 \text{Let } a &= 1, b = 4, c = 40 \\
 x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
 x &= \frac{-4 \pm \sqrt{4^2 - 4(1)(40)}}{2 \times 1} = \frac{-4 \pm \sqrt{16 - 160}}{2}
 \end{aligned}$$

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$$x = \frac{-4 \pm \sqrt{-144}}{2}$$

$$= \frac{-4 \pm 12i}{2}$$

$$= -2 \pm 6i$$

$$\therefore x = -2 + 6i, \text{ or } -2 - 6i$$

(B)

22 Given $\tan(A-B) = \frac{1}{5}$

$$\tan B = 2,$$

$$\tan A = ?$$

Recall $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

$$\frac{1}{5} = \frac{\tan A - 2}{1 + 2 \tan A}$$

$$1 + 2 \tan A = 5 \tan A - 10$$

$$5 \tan A - 2 \tan A = 1 + 10$$

$$3 \tan A = 11$$

$$\tan A = \frac{11}{3}$$

(B)

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Let $Z = 5\sqrt{3} - 5i$

$$\text{Modulus} = \sqrt{(5\sqrt{3})^2 + 5^2} \\ = \sqrt{75 + 25} = \sqrt{100} = 10 // \text{ (C)}$$

Argument of a complex number is the $\tan^{-1} \left[\frac{\text{Imaginary Part}}{\text{Real Part}} \right]$

$$\therefore \text{Arg}(Z) = \tan^{-1} \left[\frac{-5}{5\sqrt{3}} \right] = \tan^{-1} \left[-\frac{1}{\sqrt{3}} \right] = -30^\circ // \text{ (C)}$$

The polar form of a complex number of the form $Z = x + yi$ is $Z = r(\cos \theta + i \sin \theta)$

Where $r = \text{modulus} = 10$, $\theta = -30^\circ = 150^\circ [180 - 30^\circ]$

$$\therefore Z = 10(\cos 150^\circ + i \sin 150^\circ) \text{ (B)}$$

Evaluate $\frac{-1+3i}{2+i}$

$$\frac{-1+3i}{2+i} \times \frac{2-i}{2-i} = \frac{-2+i+6i+3}{4+1} = \frac{7i+1}{5} = \frac{1}{5} + \frac{7i}{5} //$$

Recall:

$$1 \text{ rad} = \frac{180^\circ}{\pi}$$

$$\therefore \frac{4\pi}{6} \text{ rad} = \frac{4\pi}{6} \times \frac{180^\circ}{\pi} = 120^\circ // \text{ (D)}$$

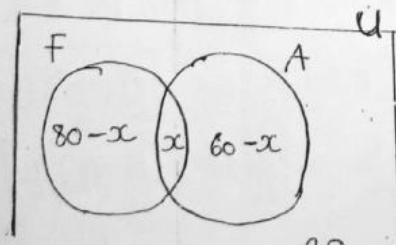
Let $U = \text{Universal Set}$

$F = \text{French}$

$A = \text{Arabic}$

$$n(U) = 150, n(F) = 80, n(A) = 60$$

$$(F \cup A)' = 20, n(F \cap A) = x$$



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$$\begin{aligned}
 80 - x + x + 60 - x + 20 &= 150 \\
 140 + 20 - x &= 150 \\
 x &= 160 - 150 = 10 // \quad (B)
 \end{aligned}$$

$$\begin{aligned}
 U &= \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14\} \\
 P &= \{2, 4, 6, 8, 10, 12, 14\} \\
 \therefore P' &= \{1, 3, 5, 7, 9, 11, 13\} \quad (C)
 \end{aligned}$$

Given $A = \{1, 2, 3, 4\}$;
 $B = \{2, 4, 6, 8\}$

The symmetric difference btw $A \Delta B = (A \cap B)'$ (C)
 $= \{1, 3, 6, 8\}$

Let $x = a$, or $x = a^2$

$$\begin{aligned}
 \therefore (x - a)(x - a^2) &= 27x^2 + bx + 8 = 0 \\
 x^2 - a^2x - ax + a^3 &= 27x^2 + bx + 8 \\
 x^2 - (a^2 + a)x + a^3 &= 27x^2 + bx + 8 \\
 x^2 - (a^2 + a)x + a^3 &= 27x^2 + bx + 8 \equiv x^2 - (a^2 + a)x + a^3 = 27x^2 + bx + 8
 \end{aligned}$$

By Comparing Coefficients,

$$a^3 = \frac{8}{27} = \left(\frac{2}{3}\right)^3$$

$$\Rightarrow a = \frac{2}{3}$$

$$-(a^2 + a) = \frac{b}{27}$$

$$-\left[\left(\frac{2}{3}\right)^2 + \frac{2}{3}\right] = \frac{b}{27} \Rightarrow \frac{b}{27} = -\left[\frac{4}{9} + \frac{2}{3}\right]$$

$$b = \frac{27 \times -10}{9} = \underline{\underline{-30}} \quad (D)$$

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$$x = \sqrt{2} + 1, \quad x = \sqrt{2} - 1$$

$$[x - (\sqrt{2} + 1)][x - (\sqrt{2} - 1)] = 0$$

$$x^2 - (\sqrt{2} - 1)x - (\sqrt{2} + 1)x + (\sqrt{2})^2 - 1^2 = 0$$

$$~~x^2 - \sqrt{2}x~~ x^2 - \sqrt{2}x + x - \sqrt{2}x - x + 2 - 1 = 0$$

$$x^2 - (1+1)\sqrt{2}x + 1 = 0$$

$$x^2 - 2\sqrt{2}x + 1 = 0$$

(A)

$$y - 11\sqrt{y} + 24 = 0$$

$$y + 24 = 11\sqrt{y}$$

$$(y + 24)^2 = (11\sqrt{y})^2 \rightarrow \text{Square both side}$$

$$y^2 + 48y + 876 = 121y$$

$$y^2 - 73y + 576 = 0$$

$$(y - 9y) - (64y + 576) = 0$$

$$y(y - 9) - 64(y - 9) = 0$$

$$(y - 9)(y - 64) = 0$$

$$\therefore y = 9 \text{ or } y = 64$$

(C)

Given: $x^2 + 2(m+1)x + (m+3)$

Using the Condition; $b^2 = 4ac$; and equating the above expression to zero (0)

$$x^2 + 2(m+1)x + (m+3) = 0; \quad a=1, \quad b=2(m+1), \quad c=m+3$$

$$[2(m+1)]^2 = 4[1 \times (m+3)]$$

$$[4[(m+1)(m+1)]] = 4m+12; \quad 4m^2+8m+4 = 4m+12$$

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$4m^2 + 4m - 8 = 0$; $m^2 + m - 2 = 0$
 Factorizing the above quadratic Equation
 $m^2 + 2m - m - 2 = 0$
 $m(m+2) - 1(m+2) = 0$
 $(m+2)(m-1) = 0$
 $\therefore m+2=0$ or $m-1=0$ $\therefore m=-2$ or $m=1$

35. $6x^2 - 10x + 4 = 0$
 $6x^2 - 6x - 4x + 4 = 0$
 $6x(x-1) - 4(x-1) = 0$
 $(6x-4)(x-1) = 0$
 Either $x-1=0$ or $6x-4=0$
 $x=1$ or $6x=4$
 $x = \frac{4}{6} = \frac{2}{3}$

5. $\sqrt{2x+9} = 1 + \sqrt{x+4}$
 Square both side
 $2x+9 = (1 + \sqrt{x+4})(x + \sqrt{x+4})$
 $2x+9 = 1 + \sqrt{x+4} + \sqrt{x+4} + x + 4$
 $2x+9 = 5 + x + 2\sqrt{x+4}$
 $2x - x + 9 - 5 = 2\sqrt{x+4}$
 $x+4 = 2\sqrt{x+4}$
 Square both side
 $(x+4)(x+4) = (2\sqrt{x+4})^2$
 $x^2 + 4x + 4x + 16 = 4(x+4)$
 $x^2 + 8x + 16 = 4x + 16$
 $x^2 + 8x - 4x = 0$
 $x^2 + 4x = 0$
 $x(x+4) = 0$
 Either $x=0$ or $x+4=0$
 $x = -4$

C
A

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$$\frac{A}{B} + \frac{A}{B} = \frac{B+A}{AB}$$

Recall $A+B = -b/a$

$$AB = c/a$$

Given $2x^2 - 6x + 5 = 0$

where $a = 2$, $b = -6$, $c = 5$

$$\therefore \frac{A+B}{AB} = \frac{-b}{a} \div \frac{c}{a} = \frac{-b}{a} \times \frac{a}{c} = \frac{-b}{c}$$

$$\therefore \frac{A+B}{AB} = \frac{-(-6)}{5} = \frac{6}{5}$$

No correct
Option!

The given sequence is an A.P., where $a =$
 $d = T_2 - T_1 = T_3 - T_2 = -3 - 1 = -4$

$$T_n = a + (n-1)d$$

Given that $T_n = -200$

$$\therefore -200 = 5 + (n-1)(-4)$$

$$-200 = 5 - 4n + 4$$

$$4n = 9 + 200$$

$$4n = 209$$

$$n = \frac{209}{4} = 52.25 \text{ term}$$

①

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39.

Given the fraction $\frac{3x-5}{(x+1)^2(x-3)^2}$
 The above partial fraction will be of the

$$\frac{3x-5}{(x+1)^2(x-3)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x-3} + \frac{D}{(x-3)^2}$$

(B)

40

Using

$$\frac{3x-5}{(x+1)^2(x-3)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x-3} + \frac{D}{(x-3)^2}$$

$$3x-5 = A(x+1)(x-3)(x-3) + B(x-3)(x-3) + C(x-3)(x+1)(x+1) + D(x+1)(x+1)$$

$$3x-5 = A(x+1)(x^2-6x+9) + B(x^2-6x+9) + C(x-3)(x^2+2x+1) + D(x^2+2x+1)$$

$$= A(x^3-6x^2+9x+x^2+9) + B(x^2-6x+9) + C(x^3+2x^2+x-3x^2-6x-3) + D(x^2+2x+1)$$

$$= A(x^3-5x^2+10x+9) + B(x^2-6x+9) + C(x^3-2x^2-5x-3) + D(x^2+2x+1)$$

By Comparing Coefficient

~~3x-5~~ For

$$x^3: A+C=0; x^2: -5A+B-C+D=0 \quad (2)$$

$$x: 3A-6B-5C+2D=3; k: 9A+9B-3C+D=-5$$

From eqn (1) $A=-C \Rightarrow$ Subt $A=-C$ into eqn 2, we have,

$$5C+B-C+D=0 \Rightarrow 4C+B+D=0 \Rightarrow B+4C+D=0$$

$$-3C-6B-5C+2D=3 \Rightarrow -6B-8C+2D=3 \quad (6) \text{ [By Subt]}$$

$$\text{Subt into eqn (4): } -9C+9B-3C+D=5 \Rightarrow 9B-12C+D=5$$

$$\text{From eqn (5) } D=-B-4C; \text{ From eqn (7) } D=-5-9B+12C$$

$$\text{From eqn (6) } D=\frac{3}{2}+3B+4C$$

Equating the D 's,

$$-B-4C=-5-9B+12C \Rightarrow 8B-16C=-5 \quad (8)$$

$$-B-4C=\frac{3}{2}+3B+4C \Rightarrow 8B+16C=-3 \quad (9)$$

Add eqn (8) & (9)

$$16B=-8 \Rightarrow B=-\frac{8}{16}=-\frac{1}{2}$$

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Put $B = -\frac{1}{2}$ into eqn (8)
 $-4 - 16C = -5 \Rightarrow 16C = -4 + 1 = -1 \Rightarrow C = -\frac{1}{16}$

Since $A = -C \Rightarrow A = \frac{1}{16}$

Put A & C into eqn $D = -B - 4C$

$$\therefore D = +\frac{1}{2} - \frac{4}{16} = \frac{1}{2} - \frac{1}{4} = \frac{2-1}{4} = \frac{1}{4}$$

Q40; $(A, B) = \left(-\frac{1}{16}, \frac{1}{2}\right)$ (No Correct Option)

$(C, D) = \left(\frac{1}{16}, \frac{1}{4}\right)$ (No Correct Option).

Given $a=5$, $d = T_2 - T_1 = T_3 - T_2 = -3 - 1 = -4$

$T_n = -75$. $n = ?$

$\therefore T_n = a + (n-1)d$

$-75 = 5 + (n-1)(-4)$

$-75 = 5 - 4n + 4$

$4n = 9 + 75$

$n = \frac{84}{4} = 21^{\text{st}} \text{ term}$

(C)

Given $T_6 = 0$, $T_{13} = 14$

$T_n = a + (n-1)d$

$T_6 = a + 5d = 0 \dots \dots \textcircled{1}$

$T_{13} = a + 12d = 14 \dots \dots \textcircled{2}$

Eqn (2) minus (-) Eqn (1)

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$$7d = 14$$

$$d = \frac{14}{7} = 2$$

Put $d = 2$ into eqn (1)

$$a = -5d = -5(2) = -10$$

For an A.P.,

First term = a

Second term = $a + d$

Third term = $a + 2d$

$\therefore$ The AP = $-10, (-10+2), (-10+2(2)), \dots$

A.P. = $-10, -8, -6, \dots$

A

Consecutive terms of an A.P. is $a-d, a$ and $a+d$.

Their sum

$$a-d + a + a+d = 18$$

$$\therefore 3a = 18$$

$$\therefore a = 6$$

Then product

$$a(a-d)(a+d) = 20$$

$$\therefore (a-d)(a+d) = 20$$

$$a^2 - d^2 = 20$$

$$d^2 = 6^2 - 20 = 36 - 20$$

$$\therefore d^2 = 16 \quad \therefore d = 4$$

$a-d$

$$a-d = 6-4 = 2$$

$$a = 6$$

$$a+d = 6+4 = 10$$

$\therefore$ the three consecutive terms

are: 2, 6, 10, —

B

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The given sequence is an A.P

$$a = -3, d = T_2 - T_1 = 2 - (-3) = 2 + 3 = 5$$

$$T_n = a + (n-1)d$$

$$\therefore T_{15} = a + 14d = -3 + 14(5) = -3 + 70 = \underline{67} \quad \textcircled{A}$$

The given sequence is a G.P

$$a = 128, T_{12} = ?, r = \frac{T_3}{T_2} = \frac{32}{64} = \frac{1}{2} \quad \textcircled{C}$$

$$T_n = ar^{n-1} \therefore T_{12} = ar^{11} = 128 \times \left(\frac{1}{2}\right)^{11} = \frac{128}{2048} = \underline{\underline{\frac{1}{16}}}$$

Given $T_4 = 24, T_8 = \frac{8}{27}$; $T_n = ar^{n-1} \rightarrow$ For G.P

$$T_4 = ar^3 = 24 \quad \text{--- (1)}$$

$$T_8 = ar^7 = \frac{8}{27} \quad \text{--- (2)}$$

Egn (2) divided by egn (1)

$$r^{7-3} = \frac{8}{27} \div 24 = \frac{8}{27} \times \frac{1}{24} = \frac{1}{84}$$

$$r^4 = \frac{1}{84} = \left(\frac{1}{3}\right)^4$$

$$\Rightarrow r = \frac{1}{3} \quad [\text{Since the power are the same}]$$

Put $r = \frac{1}{3}$ into egn (1)

$$a\left(\frac{1}{3}\right)^3 = 24 \quad \textcircled{B}$$

$$\frac{a}{27} = 24 \Rightarrow a = 24 \times 27 = \underline{\underline{648}}$$

Given $S_8 = 12, S_{16} = 56$; $S_n = \frac{n}{2}[2a + (n-1)d]$

$$S_8 = \frac{8}{2}[2a + (8-1)d] = 12$$

$$4[2a + 7d] = 12$$

$$2a + 7d = 3 \quad \text{--- (1)}$$

$$S_{16} = \frac{16}{2}[2a + (16-1)d] = 56$$

$$8[2a + 15d] = 56 \Rightarrow 2a + 15d = 7 \quad \text{--- (2)}$$

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Egn ② minus egn ①

$$8d = 4 \Rightarrow d = \frac{4}{8} = \frac{1}{2}$$

Put $d = \frac{1}{2}$ into egn ①

$$2a + 7(\frac{1}{2}) = 3$$

Multiply through by 2

$$4a + 7 = 6$$

$$4a = 6 - 7 = -1 \Rightarrow a = -\frac{1}{4}$$

For A.P, First Term = a

$$2^{\text{nd}} \text{ Term} = a + d$$

$$\text{Third Term} = a + 2d$$

$$\therefore a = -\frac{1}{4}; T_2 = -\frac{1}{4} + \frac{1}{2} = \frac{-1+2}{4} = \frac{1}{4}$$

$$T_3 = -\frac{1}{4} + 2(\frac{1}{2}) = -\frac{1}{4} + 1 = \frac{-1+4}{4} = \frac{3}{4}$$

$\therefore$ A.P is $-\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \dots$

(D)

49

The given series is a G.P

$$a = \frac{1}{8}, r = \frac{T_3}{T_2} = \frac{\frac{1}{2}}{\frac{1}{4}} = \frac{1}{2} \times \frac{4}{1} = 2; S_{10} = ?$$

$$\text{Since } r > 1, S_n = a \frac{(r^n - 1)}{r - 1}$$

$$\therefore S_{10} = \frac{1}{8} \frac{(2^{10} - 1)}{2 - 1} = \frac{1023}{8} = 127\frac{7}{8}$$

(A)

50

$$S_{\infty} = \frac{a}{1-r}; r = \frac{T_2}{T_1} = \frac{12}{20} = \frac{3}{5}; a = 2$$

$$\therefore S_{\infty} = \frac{20}{1 - \frac{3}{5}} = \frac{20}{\frac{5-3}{5}} = \frac{20 \times 5}{2} = 50$$

(B)

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$$x^2 - 3x + 1 = p(x - 3)$$

$$x^2 - x(3+p) + 1 + 3p = 0$$

$$b^2 = 4ac$$

$$(3+p)^2 = 4(1+3p)$$

$$9 + 6p + p^2 = 4 + 12p \Rightarrow 9 + 6p - 12p + p^2 - 4 = 0$$

$$p^2 - 6p + 5 = 0$$

$$p^2 - p - 5p + 5 = 0$$

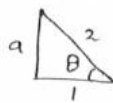
$$p(p-1) - 5(p-1) = 0$$

$$(p-5)(p-1) = 0$$

$$p = 5 \text{ or } p = 1$$



Given $\cos \theta = \frac{1}{2}$



$$a^2 = 2^2 - 1^2 = 4 - 1 = 3$$

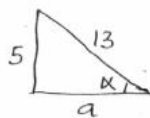
$$a = \sqrt{3}$$

$$\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{\sqrt{3}}{2}$$

$$\cot^2 \theta = \frac{\cos^2 \theta}{\sin^2 \theta} = \left(\frac{1}{2}\right)^2 \div \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{1}{4} \times \frac{4}{3} = \frac{1}{3}$$



Given $\sin x = \frac{5}{13}$



$$a^2 = 13^2 - 5^2 = 169 - 25 = 144$$

$$a = \sqrt{144} = 12$$

$$\therefore \cos x = \frac{\text{adj}}{\text{hyp}} = \frac{12}{13}$$

$$\therefore \csc^2 x = \frac{1}{\sin^2 x} = \frac{1}{\left(\frac{5}{13}\right)^2} = \frac{1}{\frac{25}{169}} = \frac{169}{25}$$



$$\cot^2 x = \frac{\cos^2 x}{\sin^2 x} = \left(\frac{12}{13}\right)^2 \div \left(\frac{5}{13}\right)^2 = \frac{144}{169} \times \frac{169}{25} = \frac{144}{25}$$



$\sec^2 x + \sin^2 x$. Recall $\sec^2 x = \frac{1}{\cos^2 x}$

$$\frac{1}{\left(\frac{12}{13}\right)^2} + \left(\frac{5}{13}\right)^2 = \frac{169}{144} + \frac{25}{169}$$



$$2\cos x + 1 = 0 \Rightarrow 2\cos x = -1$$

$$\cos x = -\frac{1}{2} \Rightarrow x = \cos^{-1}\left(-\frac{1}{2}\right) = 120^\circ$$

$$\text{In the 4th Quadrant, } x = 360 - 120 = 240^\circ$$



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58

Given $Z_1 = 2+i$; $Z_2 = 3-2i$, $Z_3 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$

$$\begin{aligned}\therefore |3Z_1 - 4Z_2| &= |3(2+i) - 4(3-2i)| \\ &= |6+3i-12+8i| \\ &= |-6+11i| \\ &= \sqrt{(-6)^2 + (11)^2} = \sqrt{36+121} = \sqrt{157}\end{aligned}$$

(B)

59

$$\begin{aligned}Z_1^3 - 3Z_1^2 + 4Z_1 - 8 \\ &= (2+i)^2(2+i) - 3(2+i)^2 + 4(2+i) - 8 \\ &= (4+4i-1)(2+i) - 3(3+4i) + 8+4i-8 \\ &= (3+4i)(2+i) - 9-12i+4i \\ &= 6+11i-4-9-8i = -7+3i\end{aligned}$$

(C)

60

$$\begin{aligned}(\bar{Z}_3)^4 &= \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^2 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^2 \\ &= \left(\frac{1}{4} - \frac{2\sqrt{3}i}{4} - \frac{3}{4}\right) \left(\frac{1}{4} - \frac{2\sqrt{3}i}{4} - \frac{3}{4}\right) \\ &= \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) \\ &= \frac{1}{4} - \frac{2\sqrt{3}i}{4} - \frac{3}{4} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i\end{aligned}$$

(C)

61

$$\begin{aligned}\left| \frac{2Z_2 + Z_1 - 5 - i}{2Z_1 - Z_2 + 3 - i} \right| &= \left| \frac{2(3-2i) + (2+i) - 5 - i}{2(2+i) - (3-2i) + 3 - i} \right| \\ &= \left| \frac{6-4i+2+i-5-i}{4+2i-3+2i+3-i} \right| = \left| \frac{3-4i}{4+3i} \right|\end{aligned}$$

NOTE

$$\frac{3-4i}{4+3i} \times \frac{4-3i}{4-3i} = \frac{12-9i-16i-12}{4^2 - (3i)^2} = \frac{-25i}{16+9} = \frac{-25i}{25}$$

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$$\therefore \left| \frac{3-4i}{4+3i} \right| = \sqrt{-1^2} = \sqrt{1} = \underline{\underline{1}} \quad \text{(A)}$$

$$2 \quad x + 2\sqrt{x} - 8 = 0$$

$$x - 8 = -2\sqrt{x}$$

Square both side

$$(x-8)^2 = (-2\sqrt{x})^2$$

$$x^2 - 16x + 64 = 4x$$

$$x^2 - 16x - 4x + 64 = 0$$

$$x^2 - 20x + 64 = 0$$

$$x(x-16) - 4(x-16) = 0$$

$$(x-4)(x-16) = 0$$

$$x = \underline{\underline{16}} \text{ or } x = \underline{\underline{4}} \quad \text{(C)}$$

$$65 \quad S_{\infty} = \frac{a}{1-r} \quad \cancel{S_n} = \frac{a(1-r^n)}{1-r}$$

$$S_3 = \frac{a(1-r^3)}{1-r} = \frac{1}{2} \cdot \frac{a}{1-r} \quad \text{(D)}$$

$$a(1-r^3) = \frac{a}{2}$$

$$1-r^3 = \frac{1}{2} \Rightarrow r^3 = 1 - \frac{1}{2} = \frac{2-1}{2} = \frac{1}{2}$$

$$r = \sqrt[3]{\frac{1}{2}} = \underline{\underline{\frac{1}{\sqrt[3]{2}}}}$$

$$3 \quad x+y=3 \quad \text{--- (1)} \quad x=3-y \quad \text{--- (2)}$$

$$y^2 + x^2 - 2y - 1 = 0$$

Recall that $(y+x)^2 = y^2 + 2xy + x^2 \Rightarrow y^2 + x^2 = (y+x)^2 - 2xy$

$$\therefore (y+x)^2 - 2xy - 2y - 1 = 0$$

$$3^2 - 2xy - 2y - 1 = 0$$

$$2xy + 2y = 8$$

$$xy + y = 4 \quad \text{--- (*)}$$

Put eqn (2) into (*)

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$$y(3-y) + y = 4$$

$$3y - y^2 + y = 4$$

$$4y - y^2 = 4$$

$$y^2 - 4y + 4 = 0$$

$$y^2 - 2y - 2y + 4 = 0$$

$$y(y-2) - 2(y-2) = 0$$

$$(y-2)(y-2) = 0$$

$$\therefore y = 2$$

Put $y = 2$ into eqn (2)

$$x = 3 - y = 3 - 2 = 1$$

$$\therefore x = 1, y = 2$$

A

64

$$\frac{3x}{(1+x)^3} = \frac{A}{1+x} + \frac{B}{(1+x)^2} + \frac{C}{(1+x)^3}$$

$$3x = A(1+x)^2 + B(1+x) + C$$

~~$$3x = A + Ax^2$$~~

$$= A(1 + 2x + x^2) + B + Bx + C$$

$$3x = A + 2Ax + Ax^2 + B + Bx + C$$

By Comparing Coefficient.

$$x^2: A = 0$$

$$x: 2A + B = 3$$

$$\Rightarrow B = 3$$

$$A + B + C = 0$$

$$0 + 3 + C = 0$$

$$C = -3$$

C

$$\therefore \frac{3}{(1+x)^3} = \frac{3}{(1+x)^2} - \frac{3}{(1+x)^3}$$

NB: No 65 is after No 62

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$$\frac{3(2^{n+1}) - 4(2^{n-1})}{2^{n+1} - 2^n} = \frac{3(2^{n+1}) - 2^{1+n}}{2^{n+1} - 2^n}$$

$$= \frac{2^{n+1}(3-1)}{2^n(2-1)} = \frac{2^{n+2}}{2^n} = 2^2 = \underline{\underline{4}} \quad \text{C}$$

$$x^3 - 2x^2 - x + 2 = 0$$

$$(x^3 - 2x^2)(-x+2) = 0$$

$$x^2(x-2) - 1(x-2) = 0$$

$$(x^2-1)(x-2) = 0$$

$$x^2 = 1 \quad \text{or} \quad x-2 = 0$$

$$x = \pm 1 \quad \text{or} \quad x = \underline{\underline{2}}$$

$$x = \pm 1 \quad \text{or} \quad \underline{\underline{2}}$$

$$\therefore x = -1, 1, \underline{\underline{2}}$$

D

$$\frac{x-1}{(x+2)(x+3)} = \frac{A}{x+2} + \frac{B}{x+3} \Rightarrow \frac{x-1}{(x+2)(x+3)} = \frac{A(x+3) + B(x+2)}{(x+2)(x+3)}$$

$$x-1 = A(x+3) + B(x+2)$$

When $x = -2$

$$-3 = A$$

$$\text{when } x = -3 \Rightarrow -B = -4 \Rightarrow B = 4$$

No Correct
Option.

$$\therefore \frac{x-1}{(x+2)(x+3)} = \frac{4}{x+3} - \frac{2}{x+2}$$

$$(2x+3y)^5 = (2x)^5 + 5(2x)^4 3y + 10(2x)^3 (3y)^2 + 10(2x)^2 (3y)^3 + 5(2x)(3y)^4 + (3y)^5$$

$$= 32x^5 + 240x^4y + 720x^3y^2 + \underline{\underline{1080x^2y^3}} + 810xy^4 + 243y^5$$

$$\therefore \text{Coefficient of } x^2y^3 = \underline{\underline{1080}} \quad \text{C}$$

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$$(1+r)^{1/3}$$

Coefficient of r^3 : ${}^nC_x \cdot a^{n-x} b^x$

$$n = \frac{1}{3}, a = 1, b = r$$

$$\therefore {}^{1/3}C_x \cdot 1^{1/3-x} r^x$$

$$\therefore \text{From } r^x = r^3 \\ \Rightarrow x = 3$$

$${}^{1/3}C_3 \cdot 1 \cdot r^3 = \frac{\frac{1}{3}!}{(\frac{1}{3}-3)! 3!} = \frac{\frac{1}{3}(\frac{1}{3}-1)(\frac{1}{3}-2)}{3 \times 2 \times 1} \cdot r^3$$

$$= \frac{5}{81} \cdot r^3$$



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A. $\tan 22.5$

from $\tan \alpha = \frac{2 \tan \frac{\alpha}{2}}{1 - \tan^2 \frac{\alpha}{2}}$

put $\alpha = 45^\circ$.

$$\tan 45 = \frac{2 \tan \left(\frac{45}{2}\right)}{1 - \tan^2 \left(\frac{45}{2}\right)} \Rightarrow \frac{2 \tan 22.5}{1 - \tan^2 22.5}$$

Let $\tan 22.5 = t$.

$$\tan 45 = \frac{2t}{1 - t^2} \quad \text{from Special angles}$$

$\tan 45^\circ = 1$.

$$1 = \frac{2t}{1 - t^2}$$

$$t^2 + 2t - 1 = 0$$

using quadratic formula,

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$t = \frac{-2 \pm \sqrt{4 - 4(1)(-1)}}{2 \cdot 1}$$

$$t = \frac{-2 \pm \sqrt{4 + 4}}{2}$$

$$t = \frac{-2 \pm \sqrt{8}}{2}$$

$$t = \frac{-2 \pm 2\sqrt{2}}{2}, \quad \therefore -1 \pm \sqrt{2}$$

Hence, $t = -1 + \sqrt{2}$ or $-1 - \sqrt{2}$

$\therefore t = \sqrt{2} - 1$ i.e. A.

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2. C

$\cos 5x + \cos x$

from trig. formulae: $\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$

Hence, $A=5x$ and $B=x$

$\therefore \cos 5x + \cos x = 2 \cos \frac{5x+x}{2} \cos \frac{5x-x}{2}$

$= 2 \cos \frac{6x}{2} \cos \frac{4x}{2} = 2 \cos 3x \cos 2x$

3. A

$\frac{10+15i}{3-4i} \times \frac{3+4i}{3+4i}$

$\frac{(10+15i)(3+4i)}{(3-4i)(3+4i)} = \frac{30+40i+45i-60}{9+16}$

$= \frac{-30+85i}{25} = \frac{-6+17i}{5}$

4. C

5. B

$y^2 = 4^x \dots (i) \therefore y = 4^{x/2} \dots (ii)$

$\therefore y^x = 16 \dots (iii) \therefore y = 16^{1/x} \dots (iv)$

equating eqn (ii) and eqn (iv)

$4^{x/2} = 16^{1/x}$

$\therefore 4^{x/2} = 4^{2/x}$

$\therefore \frac{x}{2} = \frac{2}{x}$

$\therefore x^2 = 4 \therefore x = \sqrt{4} = 2$

from eqn (ii) $y = 4^{x/2} = 4^{2/2} = 4$

$\therefore y = 4$

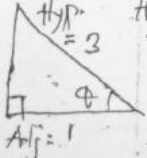
Hence, $\text{Ans} = (2, 4)$

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C $U = \{ -10, \dots, 10 \}$
 $P = \{ -9, \dots, 9 \}$, $Q = \{ -4, \dots, 3 \}$
 $P \cap Q = \emptyset$ True.
 $P \cup Q = U$ False
 $P \subset Q$ True.

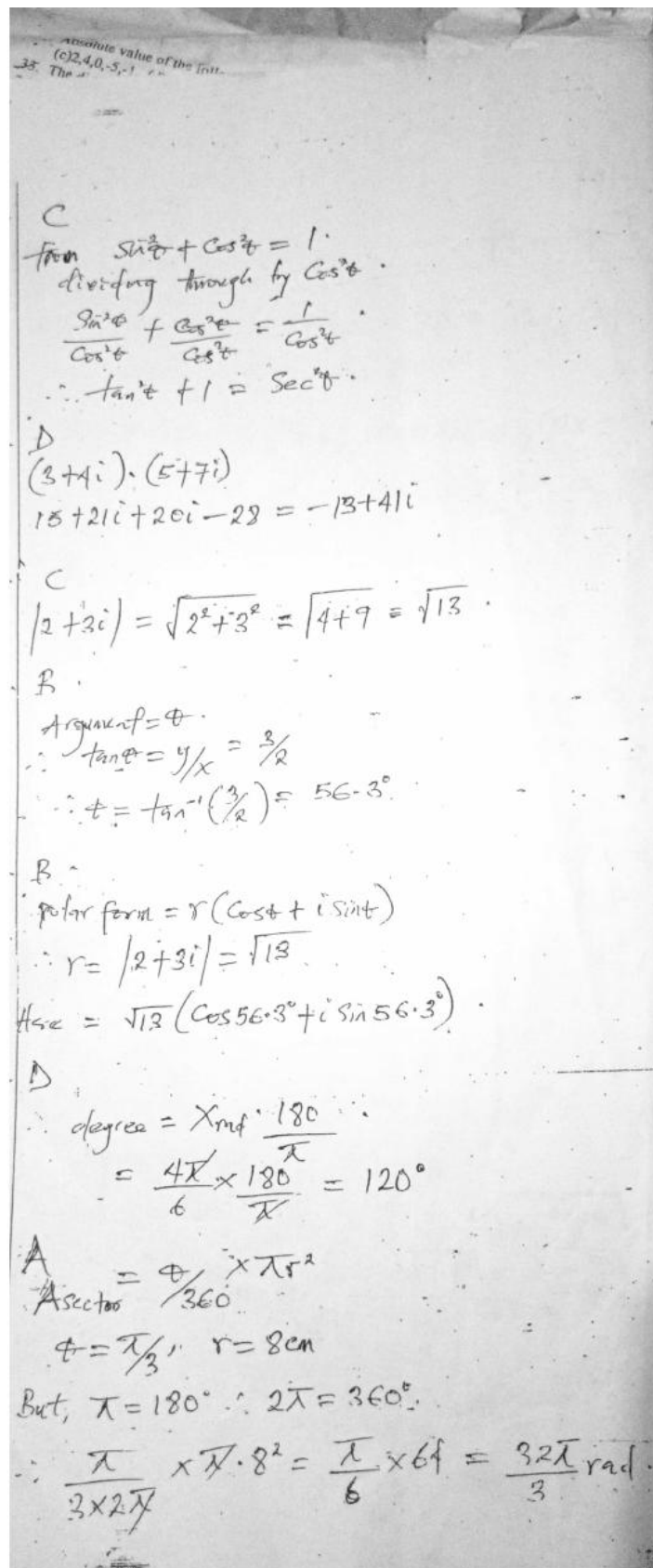
B $|3x+5| = 8$
 $(3x+5)^2 = 8^2$
 $9x^2 + 30x - 39 = 64$
 $3x^2 + 10x - 13 = 0$
 $(3x^2 + 3x) - (13x - 13) = 0$
 $3x(x+1) - 13(x+1) = 0$
 $(3x-13)(x+1) = 0$
 $\therefore x = -1$ or $3x = 13$
 $(4\frac{1}{3}, -1)$. $\therefore x = 4\frac{1}{3}$

8. D

B
 Recall $\cos \theta = \frac{\text{Adj}}{\text{Hyp}} = \frac{1}{3}$

 $\therefore \text{opp} = \sqrt{\text{Hyp}^2 - \text{Adj}^2}$
 $= \sqrt{3^2 - 1^2} = \sqrt{9-1}$
 $\therefore \text{opp} = \sqrt{8} = 2\sqrt{2}$
 $\therefore \sin \theta = \frac{\text{opp}}{\text{Hyp}} = \frac{2\sqrt{2}}{3}$

A.
 $\tan \theta = \frac{\text{opp}}{\text{Adj}} = \frac{2\sqrt{2}}{1} = 2\sqrt{2}$

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A.

$$\frac{\sqrt{2} \times \sqrt{8} \times \sqrt{32}}{\sqrt{24} \times \sqrt{26}} = \frac{\sqrt{2} \times 2\sqrt{2} \times \sqrt{3} \times \sqrt{3}}{2\sqrt{6} \times \sqrt{3} \times \sqrt{2}} = \frac{3}{\sqrt{6}}$$

A.

$$2\sqrt{8} - 3\sqrt{32} + 4\sqrt{50}$$

$$= 2\sqrt{4 \times 2} - 3\sqrt{16 \times 2} + 4\sqrt{25 \times 2}$$

$$= 2\sqrt{4} \cdot \sqrt{2} - 3\sqrt{16} \cdot \sqrt{2} + 4\sqrt{25} \cdot \sqrt{2}$$

$$= 4\sqrt{2} - 12\sqrt{2} + 20\sqrt{2}$$

$$= 12\sqrt{2}$$

A.

A.

$$A - \phi = A$$

B.

$$(A \cup B)' = A' \cap B'$$

A

$$B \cap C = \{\phi\}$$

$$\therefore A \cup (B \cap C) = A \cup \phi = A$$

$$\therefore \text{and } A = \{3, 5, 7\}$$

C

$$\frac{x^2+1}{x^3+1} = \frac{x^2+1}{(x+1)(x^2-x+1)} = \frac{A}{x+1} + \frac{Bx+C}{(x^2-x+1)} \text{ in partial}$$

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25. B

$$\frac{x}{(x+2)(x-3)} = \frac{A}{(x+2)} + \frac{B}{(x-3)}$$

$$\frac{A(x-3) + B(x+2)}{(x+2)(x-3)}$$

To find A, put $x = -2$

$$\therefore A(-2-3) = -2$$

$$\therefore -5A = -2 \therefore A = \frac{2}{5}$$

To find B, put $x = 3$

$$\therefore B(3+2) = 3 \therefore B = \frac{3}{5}$$

Hence, $\frac{x}{(x+2)(x-3)} = \frac{2}{5(x+2)} + \frac{3}{5(x-3)}$

26. B

$X = \{1, 4, 9, 16, 25, 36\}$ and $Y = \{1, 3, 5, 7, 9, 11, 13, 15\}$

$$\therefore X \cap Y = \{1, 9\}$$

27. B

$$\frac{1}{(x-3)(x-2)} = \frac{A}{(x-3)} + \frac{B}{(x-2)}$$

$$\frac{A(x-2) + B(x-3)}{(x-3)(x-2)}$$

To find A, put $x = 3$

$$A(3-2) = 1 \therefore A = 1$$

To find B, put $x = 2$

$$B(2-3) = 1 \therefore B = -1$$

$$\therefore \frac{1}{(x-3)(x-2)} = \frac{1}{x-3} - \frac{1}{x-2}$$

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D.

$$\frac{x+5}{(x-1)(x^2-1)} = \frac{x+5}{(x-1)(x-1)(x+1)}$$

$$= \frac{A}{(x-1)^2} + \frac{B}{(x-1)} + \frac{C}{(x+1)}$$

B.

$$\frac{1}{x^2-1} = \frac{1}{(x+1)(x-1)} = \frac{A}{(x+1)} + \frac{B}{(x-1)}$$

D.

A.

A.

Absolute implies: $\{5, 1, 0, 2, 4\}$.

$= 5, 1, 0, 2, 4$

Have in descending Order: $5, 4, 2, 1, 0$.

B.

C.

C.

As natural numbers, whole numbers, integers and complex numbers
a subset of rational numbers which is a set of real numbers.

A.

C.

from $\text{degree} = X_{\text{rad}} \cdot \frac{180}{\pi}$

$$\therefore X_{\text{rad}} = \frac{\text{degree} \times \pi}{180} = \left(\frac{35 \times \pi}{180} \right) \text{rad}$$

$$= \frac{7\pi}{36} \text{rad}$$

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32. B.

Using Heron's formula,

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{where } s = \frac{a+b+c}{2}$$

$$\therefore \text{given: } a=10\text{cm}, b=7\text{cm}, c=5\text{cm}$$

$$\therefore s = \frac{10+7+5}{2} = 11\text{cm}$$

$$\therefore A = \sqrt{11(11-10)(11-7)(11-5)}$$

$$\therefore A = \sqrt{264} = 16.25\text{cm}^2$$

39. A.

$$\tan 75^\circ = \tan (45^\circ + 30^\circ)$$

$$\text{Recall, } \tan (A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

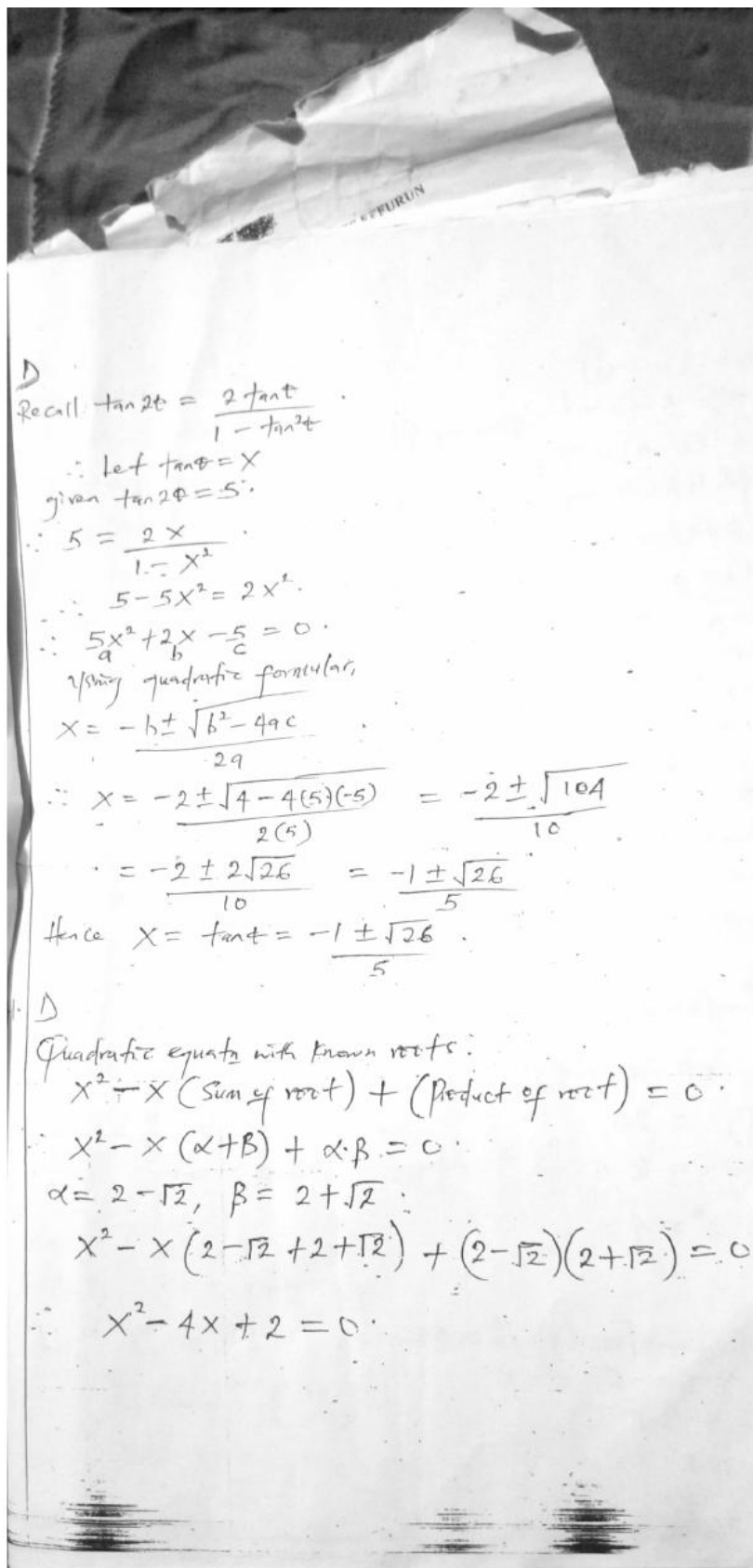
$$\therefore \tan (45^\circ + 30^\circ) = \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ}$$

$$\text{Recall from Special angles, } \tan 45^\circ = 1; \tan 30^\circ = \frac{\sqrt{3}}{3}$$

$$\therefore \tan 75^\circ = \frac{1 + \frac{\sqrt{3}}{3}}{1 - 1 \cdot \frac{\sqrt{3}}{3}} = \frac{3 + \sqrt{3}}{3 - \sqrt{3}} \times \frac{3 + \sqrt{3}}{3 + \sqrt{3}}$$

$$\therefore \tan 75^\circ = \frac{12 + 6\sqrt{3}}{6} = 2 + \sqrt{3}$$

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42. B

$$m - n = 3 \quad \text{--- (i)}$$

$$m^2 - n^2 = 9 \quad \text{--- (ii) from eqn (i)}$$

$$(m+n)(m-n) = 9 \quad \text{--- (iii)}$$

Substitute eqn (i) into eqn (iii)

$$3(m+n) = 9$$

$$\therefore m+n = 3 \quad \text{--- (iv)}$$

$$\begin{array}{r} m-n = 3 \quad \text{--- (i)} \\ m+n = 3 \quad \text{--- (iv)} \end{array} \quad \begin{array}{l} \text{By Elimination} \\ \text{Method} \end{array}$$

Adding both eqns

$$m+n = 3+3$$

$$\therefore 2m = 6$$

$$\therefore m = 3$$

Substitute $m=3$ into eqn (i)

$$3 - n = 3 \quad \therefore n = 3 - 3 = 0$$

$$\therefore (m, n) = (3, 0)$$

43. D

$$(\sqrt{y+5})^2 = (5-\sqrt{y})^2$$

$$y+5 = 25 - 10\sqrt{y} + y$$

$$\therefore 10\sqrt{y} = 20$$

$$\therefore \sqrt{y} = 2$$

$$\therefore y = 2^2 = 4$$

44. C

Condition for equal roots: $b^2 = 4ac$

$$\therefore a = 3, b = -6, c = K$$

$$(-6)^2 = 4 \cdot 3 \cdot K$$

$$36 = 12K$$

$$\therefore K = \frac{36}{12} = 3$$

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A.

Interpretation: $2n^2 = 3n + 5$.

$$\therefore 2n^2 - 3n - 5 = 0$$

$$\therefore (2n^2 + 2n) - (5n - 5) = 0$$

$$\therefore 2n(n+1) - 5(n+1) = 0$$

$$\therefore (n+1)(2n-5) = 0$$

$$\therefore n = -1, \quad 2n - 5 = 0$$

$$2n = 5 \quad \therefore n = \frac{5}{2} = 1\frac{1}{2}$$

$$= (-1, \frac{5}{2})$$

C

Discriminant $D = b^2 - 4ac$

$D \geq 0$, implies the roots are real.

$D = 0$, implies the roots are equal

$D > 0$, implies the roots are real and different.

$D < 0$, implies the roots are Imaginary Or Complex.

A.

$a = 3, b = 5, c = -2$.

recall, $\alpha + \beta = -b/a = -5/3$.

$\alpha \cdot \beta = c/a = -2/3$.

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha \cdot \beta} = \frac{-5/3}{-2/3} = \frac{5}{2}$$

A.

as $r = T-1$

$T = 3, \therefore r = 3-1 = 2$.

$T_{r+1} = {}^nC_r \cdot a^{n-r} \cdot b^r$ as $a = 2x, b = \frac{3y}{4}, n = 3$.

$${}^3C_2 \cdot (2x)^{3-2} \cdot \left(\frac{3y}{4}\right)^2$$

$$3 \cdot 2 \cdot x \cdot \frac{9}{16} \cdot y^2 = \frac{27}{8} xy^2$$

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49. B.

$$\text{From } T_{r+1} = {}^nC_r \cdot a^{n-r} \cdot b^r$$

$$n=6, a=1, b=-2y.$$

$$\therefore {}^6C_r \cdot (1)^{6-r} \cdot (-2y)^r = {}^6C_r \cdot (1)^6 \cdot (-2)^r (y)^r \cdot (y^0)^{-r}$$

$$\therefore y^r = y^4 \quad \therefore r=4.$$

$$\therefore {}^6C_4 (1)^{6-4} (-2)^4 y^4 = 15 \cdot 1 \cdot 16 \cdot y^4 = 240$$

50. B.

$$\text{Recall, } T_{r+1} = {}^nC_r \cdot a^{n-r} \cdot b^r.$$

$$\therefore \text{General term } T_{r+1} = {}^nC_r \cdot (1)^{n-r} \cdot (2)^r \cdot y^r$$

$$\therefore x + y^4 = y^r \text{ by Comparison.}$$

$$\therefore r=4$$

$$\therefore T_{r+1} = {}^nC_4 \cdot (1)^{n-4} \cdot 2^4 \cdot y^4.$$

$$\text{Hence, Coeff of } y^4 = \frac{n!}{(n-4)!4!} \cdot 16.$$

$$\text{at } y^3 = y^r \text{ by Comparison.}$$

$$\therefore r=3.$$

$$\therefore T_{r+1} = {}^nC_3 \cdot (1)^{n-3} \cdot 2^3 \cdot y^3$$

$$\text{Hence, Coeff of } y^3 = \frac{n!}{(n-3)!3!} \cdot 8.$$

$$\text{But, } \frac{n! \cdot 16 / n! \cdot 8}{(n-4)!4! / (n-3)!3!} = 3/1$$

$$\therefore \frac{n! \cdot 2 \cdot 16 / n! \cdot 8}{(n-4)!4 \times 3! / (n-3)(n-4)!3!} = 3/1$$

$$\therefore \frac{n-3}{2} = \frac{3}{1}$$

$$n-3 = 6$$

$$\therefore n = 6+3 = 9.$$

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coefficient of the fourth 4th power using Pascal's triangle:

$$1 \quad 4 \quad 6 \quad 4 \quad 1$$

$$(\sqrt{3})^4 = 1 + 4\sqrt{3} + 6(\sqrt{3})^2 + 4(\sqrt{3})^3 + (\sqrt{3})^4$$

$$= 1 + 4\sqrt{3} + 18 + 12\sqrt{3} + 9 = 28 + 16\sqrt{3}$$

$$(-\sqrt{3})^4 = 1 + 4(-\sqrt{3}) + 6(-\sqrt{3})^2 + 4(-\sqrt{3})^3 + (-\sqrt{3})^4$$

$$= 1 - 4\sqrt{3} + 18 - 12\sqrt{3} + 9 = 28 - 16\sqrt{3}$$

$$(1+\sqrt{3})^4 - (1-\sqrt{3})^4 = (28 + 16\sqrt{3}) - (28 - 16\sqrt{3})$$

$$= 28 + 16\sqrt{3} - 28 + 16\sqrt{3} = 32\sqrt{3}$$

B.

Recall, $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{6}x^3 + \dots$

From $(1+x)^{1/2}$, $\therefore n = 1/2$

Coef of $x^3 = \frac{n(n-1)(n-2)}{6} = \frac{1/2(1/2-1)(1/2-2)}{6}$

$$= 1/16$$

Recall $T_n = S_n - S_{n-1}$

$$\therefore T_4 = S_4 - S_3$$

But $S_n = 2n(n+3)$

$$\therefore S_4 = 2 \cdot 4 \cdot (4+3) = 56$$

$$S_3 = 2 \cdot 3 \cdot (3+3) = 36$$

$$\therefore T_4 = 56 - 36$$

$$= 20$$

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56. C

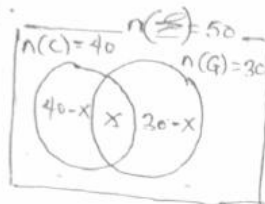
From $A = \{2, 9, 5, 2\} \therefore n(A) = 3$

$$P = 2^n = 2^3 = 8 \text{ as I don't count a repeated element again.}$$

57. A

58. D

59. A.



$$\therefore 40 - X + X + 30 - X = 50$$

$$70 - X = 50.$$

$$\therefore X = 70 - 50 = 20.$$

$$\therefore n(C \cap G) = 20.$$

60. C

$$X = \{1, 5, 10\}, Y = \{2, 4, 6\}$$

$$\therefore X \cap Y = \emptyset$$

61. A.

$$a = T_1 = -\frac{1}{2}$$

$$d = T_2 - T_1 = T_3 - T_2 = \frac{3}{2} + \frac{1}{2} = 2.$$

$$\text{Recall, } T_n = a + (n-1)d.$$

$$\therefore T_n = -\frac{1}{2} + 2(n-1) = 2n - \frac{5}{2}$$

$$\therefore T_n = \frac{1}{2}(4n - 5).$$

62. B

$$T_3 = a + 2d = -5 \quad \text{--- (i)}$$

$$T_5 = a + 4d = 4.5 \quad \text{--- (ii)}$$

Subtracting eqn (i) from eqn (ii) =

$$4d - 2d = 4.5 + 5$$

$$\therefore 2d = 9.5$$

$$\therefore d = 4.75$$

- from eqn (i), $a + 2d = -5$

$$\therefore a = -5 - 2d = -5 - 2(4.75)$$

$$\therefore a = -5 - 9.5$$

$$\therefore a = -14.5.$$

$$\therefore T_{13} = a + 12d = -14.5 + 12(4.75)$$

$$= 57 - 14.5$$

$$T_{13} = 42.5$$

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3. C

$$T_1 = a = 19 \quad \text{--- (i)}$$

$$T_4 = a + 3d = -8 \quad \text{--- (ii)}$$

Substitute eqn (i) into eqn (ii)

$$19 + 3d = -8$$

$$3d = -27 \therefore d = -9$$

$$X = T_2 = a + d = 19 - 9 = 10$$

$$Y = T_3 = a + 2d = 19 - 18 = 1$$

$$\therefore (X, Y) = (10, 1)$$

D

$$a = 1\frac{1}{2} = \frac{3}{2}$$

$$d = T_2 - T_1 = \frac{5}{4} - \frac{3}{2} = -\frac{1}{4}$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$\therefore S_{20} = \frac{20}{2} [2 \cdot \frac{3}{2} + (20-1) \cdot (-\frac{1}{4})] = 30 - 47.5$$

$$\therefore S_{20} = -17.5$$

5. A

$$S_n = 252, a = -16, l = 72$$

Recall, $S_n = \frac{n}{2} [a + L]$

$$\frac{n}{2} [-16 + 72] = 252$$

$$\therefore n = 9$$

B

Recall for G.P. $T_n = ar^{n-1}$

$$\therefore T_1 = a = 81$$

$$T_5 = ar^4 = 1$$

$$\therefore \frac{T_5}{T_1} = \frac{ar^4}{a} = \frac{1}{81} \therefore r = \frac{1}{3}$$

$$\therefore T_7 = ar^6 = 81 \left(\frac{1}{3}\right)^6 = \frac{1}{9}$$

C

$$a = 1, r = t$$

$$S_n = \frac{a}{1-r} = \frac{1}{1-t}$$

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68. D

$$a = 2, d = n+2.$$

$$\text{Recall, } T_n = a + (n-1)d.$$

$$\therefore T_n = 2 + (n-1)(n+2)$$

$$\therefore T_n = 2 + n^2 + n - 2$$

$$\therefore T_n = n^2 + n.$$

$$\text{Hence, } T_n = n(n+1).$$

69. C

$$\text{From: } T_{n+1} + n^2 = nT_n + 2$$

$$\text{at } n=1, T_1 = 2.$$

$$\therefore T_{1+1} + 1^2 = 1 \cdot T_1 + 2$$

$$\therefore T_2 + 1 = T_1 + 2.$$

$$\therefore T_2 = T_1 + 2 - 1 = 2 + 2 - 1$$

$$\therefore T_2 = 4 - 1 = 3.$$

$$\text{at } n=2.$$

$$T_{2+1} + 2^2 = 2 \cdot T_2 + 2$$

$$T_3 + 4 = 2T_2 + 2$$

$$\therefore T_3 = 2 \cdot 3 - 4 = 4.$$

$$\therefore T_3 = 4.$$

$$\therefore (T_2, T_3) = (3, 4)$$

70. D

From intuition, $x+1$ is a factor
 $x^3 - 2x^2 + x + 2$.

$$\text{Hence, } \frac{x^3 - x^2 + 2}{x+1}$$

$$x+1 \overline{) x^3 - 2x^2 + x + 2}$$

$$\underline{x^3 + x^2}$$

$$-x^2 + x + 2$$

$$\underline{-x^2 - x}$$

$$2x + 2$$

$$\underline{2x + 2}$$

$$\text{factors: } (x+1)(x^2 - x + 2).$$

But $(x^2 - x + 2)$ cannot be factor.
 Hence, the problem assumes

$$x^2 - x - 2 = 0$$

$$(x^2 + x) - (2x - 2) = 0$$

$$x(x+1) - 2(x-1) = 0$$

$$(x+1)(x-2) = 0.$$

$$\text{Hence, factors: } (x+1)(x+1)(x-2)$$

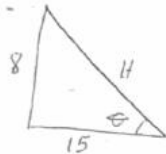
$$\therefore (-1, -1, 2)$$

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ANSWERS TO MTH 111 - 2015/2016 TYPE D

① D

② Given $\tan \theta = \frac{8}{15}$



Applying the pythagoras theorem:

$$H^2 = 8^2 + 15^2$$

$$H = \sqrt{64 + 225} = \sqrt{289} = 17$$

$$\sin \theta = \frac{8}{17}, \quad \cos \theta = \frac{15}{17}$$

$$\frac{\sin 2\theta}{\cos \theta - \sin \theta} = \frac{2 \sin \theta \cos \theta}{\cos \theta - \sin \theta} = \left(2 \times \frac{8}{17} \times \frac{15}{17} \right) \div \left(\frac{15}{17} - \frac{8}{17} \right)$$

$$= \frac{240}{289} \div \left(\frac{15-8}{17} \right)$$

$$= \frac{240}{289} \times \frac{17}{7} = \frac{240}{119}$$

(A)

③ Given $\cos 2A = 2 \cos^2 A - 1$

$$\cos 4A = \cos (2A + 2A)$$

$$\text{Recall that: } \cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos(2A + 2A) = \cos 2A \cos 2A - \sin 2A \sin 2A$$

$$= \cos^2 2A - \sin^2 2A$$

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Recall that, $\sin 2A = 2 \sin A \cos A$

$$\begin{aligned} \therefore \cos 4A &= (2 \cos^2 A - 1)(2 \cos^2 A - 1) - [(2 \sin A \cos A)(2 \sin A \cos A)] \\ &= 4 \cos^4 A - 2 \cos^2 A - 2 \cos^2 A + 1 - [4 \sin^2 A \cos^2 A] \\ &= 4 \cos^4 A - 4 \cos^2 A + 1 - 4(1 - \cos^2 A) \cos^2 A \\ &= 4 \cos^4 A - 4 \cos^2 A + 1 - 4 \cos^2 A + 4 \cos^4 A \\ &= 8 \cos^4 A - 8 \cos^2 A + 1 \quad \text{--- (A)} \end{aligned}$$

From $\sin A + \sin B = p$

$$\cos A + \cos B = q$$

But $\sin A + \sin B = 2 \sin \frac{1}{2}(A+B) \cos \frac{1}{2}(A-B)$

$$\cos A + \cos B = 2 \cos \frac{1}{2}(A+B) \cos \frac{1}{2}(A-B)$$

$$\therefore 2 \sin \frac{1}{2}(A+B) \cos \frac{1}{2}(A-B) = p \quad \text{--- (1)}$$

$$2 \cos \frac{1}{2}(A+B) \cos \frac{1}{2}(A-B) = q \quad \text{--- (2)}$$

Squaring both equations, then add them:

$$\begin{aligned} p^2 + q^2 &= 4 \cos^2 \frac{1}{2}(A-B) [\sin^2 \frac{1}{2}(A+B) + \cos^2 \frac{1}{2}(A+B)] \\ \cos^2 \frac{1}{2}(A-B) &= \frac{p^2 + q^2}{4} \end{aligned}$$

$$\begin{aligned} \text{Recall: } \sin^2 \frac{1}{2}(A+B) + \cos^2 \frac{1}{2}(A+B) &= 1 \\ \sin^2 \theta + \cos^2 \theta &= 1 \end{aligned}$$

$$\text{But } \sin^2 \frac{1}{2}(A-B) = 1 - \cos^2 \frac{1}{2}(A-B)$$

$$\begin{aligned} &= 1 - \frac{p^2 + q^2}{4} \\ &= \frac{4 - (p^2 + q^2)}{4} \end{aligned}$$

$$\tan^2 \frac{1}{2}(A-B) = \frac{4 - (p^2 + q^2)}{4} \times \frac{4}{p^2 + q^2} \quad \left[\tan^2 \theta = \frac{\sin^2 \theta}{\cos^2 \theta} \right]$$

$$= \frac{4}{p^2 + q^2} - 1 \Rightarrow 4(p^2 + q^2)^{-1} - 1 \quad \text{--- (A)}$$

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⑤ $\sin 16x + \sin 14x$

Recall: $\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$

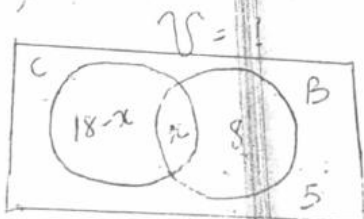
$$\begin{aligned}\therefore \sin 16x + \sin 14x &= 2 \sin\left(\frac{16x+14x}{2}\right) \cos\left(\frac{16x-14x}{2}\right) \\ &= 2 \sin\left(\frac{30x}{2}\right) \cos\left(\frac{2x}{2}\right) \\ &= 2 \sin 15x \cos x \quad (A)\end{aligned}$$

⑥ $\cos 18x - \cos 12x$

Recall: $\cos A - \cos B = -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$

$$\begin{aligned}\therefore \cos 18x - \cos 12x &= -2 \sin\left(\frac{18x+12x}{2}\right) \sin\left(\frac{18x-12x}{2}\right) \\ &= -2 \sin\left(\frac{30x}{2}\right) \sin\left(\frac{6x}{2}\right) \\ &= -2 \sin 15x \sin 3x \quad (A)\end{aligned}$$

$n(C) = 18$, $n(B) = 22$, $n(C \cup B)' = 5$
 $n(B) \cap C = 8$



$$8 + x = 22$$

$$x = 22 - 8 = 14$$

$$\begin{aligned}n(U) &= 18 - x + x + 8 + 5 \\ &= 31\end{aligned}$$

⑧ From Q7

$$\begin{aligned} n(\text{Smoke without drinking}) &= 18 - x \\ &= 18 - 14 \\ &= 4 \end{aligned}$$

(B)

2. If $A \subset B$, then $B^c \subset A^c$

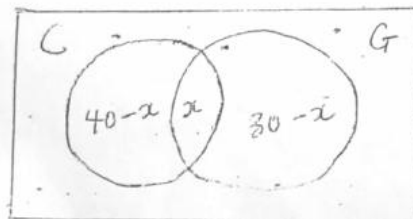
$$\text{Recall: } B^c \equiv A, A^c \equiv B$$

(D)

Let $C = \text{Champion}$, $G = \text{Guardian}$

$$n(U) = 50, \quad n(C) = 40, \quad n(G) = 30, \quad n(C \cap G) = x$$

$$U = 50$$



$$U = 40 - x + x + 30 - x$$

$$50 = 70 - x$$

$$x = 70 - 50 = 20$$

Number of people that read both = 20 people (A)

$$T_3 = -9, \quad T_7 = -29, \quad a = ?, \quad d = ?, \quad T_{10} = ?$$

$$T_n = a + (n-1)d$$

$$T_3 = a + 2d = -9 \quad \text{--- (1)}$$

$$T_7 = a + 6d = -29 \quad \text{--- (2)}$$

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Substn eqn ① from eqn ②

$$4d = -20$$

$$d = \frac{-20}{4} = -5$$

Put $d = -5$ into eqn (1)

$$a + 2(-5) = -9$$

$$a = -9 + 10 = 1$$

$$\begin{aligned} T_{10} &= a + 9d = 1 + 9(-5) \\ &= 1 - 45 \\ T_{10} &= -44 \quad (C) \end{aligned}$$

$$(12) \quad a = 3, T_2 = 6, T_3 = 9$$

$$d = T_3 - T_2 = 9 - 6 = 3$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{18} = \frac{18}{2} [2(3) + (18-1)3]$$

$$= 9 [6 + 17(3)] = 9 [6 + 51]$$

$$= 9(57) = 513$$

$$S_{18} = 513 \quad (B)$$

$$(13) \quad C \rightarrow (A-B) \cup (B-A)$$

$$(14) \quad D \rightarrow A \cap B^c$$

$$C = \{1, 2, 3, 4\}, \quad D = \{2, 4, 6, 8\}$$

$$\text{Symmetric difference} = (C - D) \cup (D - C)$$

$$\text{Where } C - D = \{1, 3\}$$

$$D - C = \{6, 8\}$$

$$\therefore (C - D) \cup (D - C) = \{1, 3, 6, 8\} \quad (A)$$

$$C \rightarrow \{\dots, -4, 0, 4, \dots\}$$

$$x^2 + (m-2)x + m+1 = 0$$

$$\text{Condition for equal roots} \Rightarrow b^2 = 4ac$$

$$\therefore (m-2)^2 = 4(1)(m+1)$$

$$(m-2)(m-2) = 4(m+1)$$

$$m^2 - 2m - 2m + 4 = 4m + 4$$

$$m^2 - 4m + 4 = 4m + 4$$

$$m^2 - 4m - 4m = 0$$

$$m^2 - 8m = 0$$

$$m(m-8) = 0$$

$$m = 0 \quad \text{or} \quad m-8 = 0 \quad , \quad m = 8$$

$$\therefore m = 0 \quad \text{or} \quad 8$$

$$m = (0, 8)$$

NO OPTIO

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18

$$\sqrt{a+5} = 5-\sqrt{a}$$

Square both sides

$$(\sqrt{a+5})^2 = (5-\sqrt{a})^2$$

$$a+5 = (5-\sqrt{a})(5-\sqrt{a})$$

$$a+5 = 5(5-\sqrt{a}) - \sqrt{a}(5-\sqrt{a})$$

$$a+5 = 25 - 5\sqrt{a} - 5\sqrt{a} + a$$

$$5 = 25 - 10\sqrt{a}$$

$$10\sqrt{a} = 25 - 5$$

$$10\sqrt{a} = 20$$

$$\sqrt{a} = \frac{20}{10} = 2$$

$$(\sqrt{a})^2 = 2^2$$

$$\therefore a = 4 \quad (C)$$

19

$$x^2 + 2x + 1 = 0$$

$$x^2 + 2x + 1 = 0$$

$$\alpha + \beta = \frac{-b}{a} = \frac{-4}{2} = -2$$

$$\alpha\beta = \frac{c}{a} = \frac{7}{2}$$

$$\alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2)$$

$$= (\alpha + \beta)(\alpha^2 + \beta^2 - \alpha\beta)$$

Recall: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$$= (\alpha + \beta)((\alpha + \beta)^2 - 3\alpha\beta)$$

$$= -2((-2)^2 - 3(\frac{7}{2}))$$

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$$\begin{aligned}
 &= -2\left(4 - \frac{21}{2}\right) \\
 &= -2\left(\frac{8-21}{2}\right) \\
 &= -1(-13) = 13 \quad (B)
 \end{aligned}$$

Let the age of Toju be x
 and the age of Misan be y

$$x - y = 6 \quad \text{--- (1)}$$

$$xy = 187 \quad \text{--- (2)}$$

From eqn (1), $x = 6 + y$ --- (3)

Put eqn (3) into eqn (2)

$$(6 + y)y = 187$$

$$6y + y^2 = 187$$

$$y^2 + 6y - 187 = 0$$

Using quadratic formula:

$$a = 1, \quad b = 6, \quad c = -187$$

$$y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-6 \pm \sqrt{6^2 - 4(1)(-187)}}{2(1)}$$

$$y = \frac{-6 \pm \sqrt{36 + 748}}{2} = \frac{-6 \pm \sqrt{784}}{2}$$

$$= \frac{-6 \pm 28}{2}$$

$$y = \frac{-6 + 28}{2} \quad \text{or} \quad \frac{-6 - 28}{2}$$

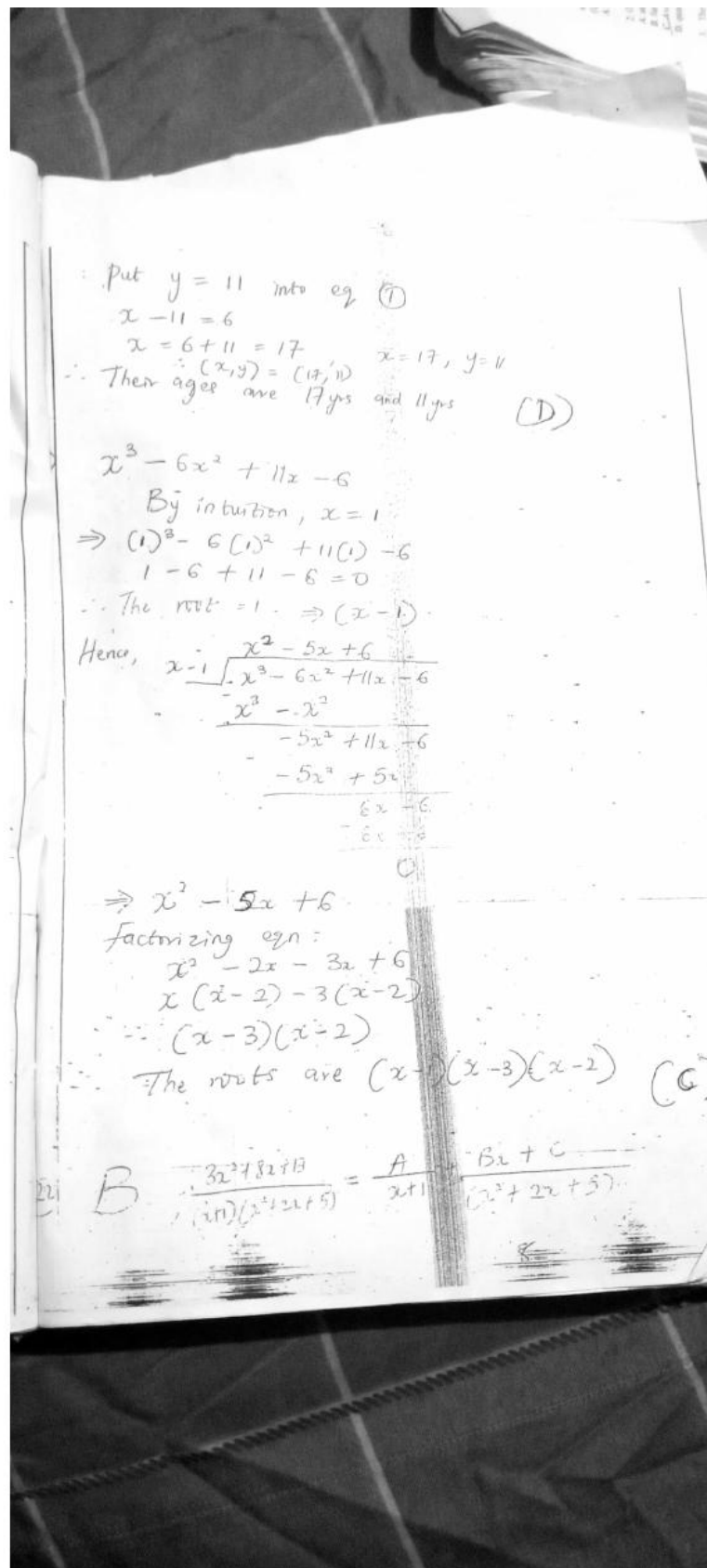
$$= \frac{22}{2} \quad \text{or} \quad \frac{-34}{2}$$

$$y = 11 \quad \text{or} \quad -17$$

Age can never have a negative sign.

$$y = 11$$

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$$\frac{3x^2 + 8x + 13}{(x+1)(x^2+2x+5)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+2x+5}$$

$$\begin{aligned} 3x^2 + 8x + 13 &= A(x^2 + 2x + 5) + (Bx + C)(x + 1) \\ &= Ax^2 + 2Ax + 5A + Bx^2 + Cx + Bx + C \\ &= Ax^2 + 2Ax + 5A + Bx^2 + Cx + Bx + C \end{aligned}$$

$$3x^2 + 8x + 13 = x^2(A+B) + x(2A+B+C) + 5A+C$$

When $x = -1$

$$3(-1)^2 + 8(-1) + 13 = A + B - 2A - B - C + 5A + C$$

$$3 - 8 + 13 = 4A$$

$$8 = 4A$$

$$A = \frac{8}{4} = 2$$

$$\therefore A = 2$$

NO OPTION

Comparing Coefficients;

$$A + B = 3 \quad \text{--- (1)}$$

$$2A + B + C = 8 \quad \text{--- (2)}$$

$$5A + C = 13 \quad \text{--- (3)}$$

from eq (3), put $A = 2$

$$5(2) + C = 13$$

$$C = 13 - 10 = 3$$

$$\therefore C = 3$$

From eq. (1); put $A = 2$

$$2 + B = 3$$

$$\therefore B = 3 - 2 = 1$$

$$\therefore C = 3, B = 1$$

$$(A, B) = (2, 1)$$

(D)

$$x - 8\sqrt{x} + 15 = 0$$

$$x + 15 = 8\sqrt{x}$$

Square both sides

$$(x + 15)^2 = (8\sqrt{x})^2$$

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$$\begin{aligned}
 x^2 + 30x + 225 &= 64x \\
 x^2 - 34x + 225 &= 0 \\
 x^2 - 9x - 25x + 225 &= 0 \\
 x(x-9) - 25(x-9) &= 0 \\
 (x-25)(x-9) &= 0 \\
 \text{Either } x &= 25 \text{ or } x = 9 \Rightarrow (25, 9) = (x, y) \quad (A)
 \end{aligned}$$

$$\begin{aligned}
 x + y &= 1 \quad \text{--- (1)} \\
 x^2 + y^2 &= 13 \quad \text{--- (2)} \\
 x &= 1 - y \quad \text{--- (3)} \\
 \text{Put } x &= 1 - y \text{ into eqn (2)} \\
 (1 - y)^2 + y^2 &= 13 \\
 1 - 2y + y^2 + y^2 &= 13 \\
 2y^2 - 2y + 1 - 13 &= 0 \\
 2y^2 - 2y - 12 &= 0 \\
 y^2 - y - 6 &= 0 \\
 y^2 - 3y + 2y - 6 &= 0 \\
 y(y-3) + 2(y-3) &= 0 \\
 (y+2)(y-3) &= 0 \\
 y &= -2 \quad \text{or} \quad y = 3 \\
 \text{Put } y_1 &= -2 \text{ and } y_2 = 3 \text{ into eq (1)} \\
 x + y_1 &= 1 \quad \text{or} \quad x + y_2 = 1 \\
 x + (-2) &= 1 \quad \text{or} \quad x + 3 = 1 \\
 x &= 1 + 2 \quad \text{or} \quad x = 1 - 3 \\
 x_1 &= 3 \quad \text{or} \quad x_2 = -2 \\
 \therefore (x_1, y_1) &= (3, -2) \text{ and } (y_1, y_2) = (-2, 3)
 \end{aligned}$$

$$\begin{aligned}
 \tan 105^\circ &\equiv \tan (45^\circ + 60^\circ) \\
 \text{Recall } \tan (A+B) &= \frac{\tan A + \tan B}{1 - \tan A \tan B} \\
 \tan 105^\circ &= \frac{\tan 45^\circ + \tan 60^\circ}{1 - \tan 45^\circ \tan 60^\circ}
 \end{aligned}$$

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$$\begin{aligned}
 \therefore \tan 105 &= \frac{1+\sqrt{3}}{1-i(\sqrt{3})} = \frac{1+\sqrt{3}}{1-\sqrt{3}} \\
 &= \frac{1+\sqrt{3}}{1-\sqrt{3}} \times \frac{1+\sqrt{3}}{1+\sqrt{3}} \\
 &= \frac{1+2\sqrt{3}+3}{1^2-(\sqrt{3})^2} \\
 &= \frac{4+2\sqrt{3}}{1-3} = \frac{4+2\sqrt{3}}{-2}
 \end{aligned}$$

$$\therefore \tan 105 = -2 - \sqrt{3} \quad (A)$$

$$\cos 255 = \cos (180^\circ + 75^\circ) = \cos 180^\circ \cos 75^\circ - \sin 180^\circ \sin 75^\circ$$

$$\text{Recall: } = -1 \times \cos(30^\circ + 45^\circ) - 0 \times \sin 75^\circ$$

$$= -1 [\cos 30^\circ \cos 45^\circ - \sin 30^\circ \sin 45^\circ]$$

$$= -1 \left[\frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} - \frac{1}{2} \times \frac{1}{\sqrt{2}} \right]$$

$$= -1 \left(\frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} \right)$$

$$= -1 \left(\frac{\sqrt{3}-1}{2\sqrt{2}} \right)$$

$$= \frac{-\sqrt{3}+1}{2\sqrt{2}} \times \frac{2\sqrt{2}}{2\sqrt{2}}$$

$$= \frac{-2\sqrt{6}+2\sqrt{2}}{4\sqrt{2}} = \frac{-\sqrt{6}+\sqrt{2}}{2}$$

$$= \frac{-\sqrt{6}+\sqrt{2}}{2}$$

$$\therefore \cos 255 = \frac{1}{4} (\sqrt{2} - \sqrt{6}) \quad (A)$$

$$\tan 3A = \tan (A + 2A)$$

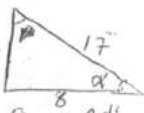
$$= \frac{\tan A + \tan 2A}{1 - \tan A \tan 2A}$$

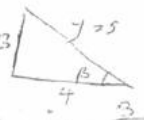
$$1 - \tan A \tan 2A$$

$$\text{Recall: } \tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

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$$\begin{aligned}
 &= \frac{\tan A + \frac{2 \tan A}{1 - \tan^2 A}}{1 - \left(\tan A - \frac{2 \tan A}{1 - \tan^2 A} \right)} \\
 &= \frac{\tan A (1 - \tan^2 A) + 2 \tan A}{1 - \tan^2 A} \div \frac{(1 - \tan^2 A - 2 \tan^2 A)}{1 - \tan^2 A} \\
 &= \frac{\tan A - \tan^3 A + 2 \tan A}{1 - \tan^2 A} \times \frac{1 - \tan^2 A}{1 - 3 \tan^2 A} \\
 &= \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A} \quad (B)
 \end{aligned}$$

$15 = x$

 $\cos \alpha = \frac{8}{17} = \frac{\text{adj}}{\text{hyp}}$
 $x^2 = 17^2 - 8^2$
 $= 289 - 64 = 225$
 $= \sqrt{225} = 15$
 $\sin \alpha = \frac{15}{17}, \tan \alpha = \frac{15}{8}$


 $\tan \beta = \frac{3}{4} = \frac{\text{opp}}{\text{adj}}$
 $y^2 = 3^2 + 4^2$
 $= 9 + 16$
 $= 25 = 5$
 $\cos \beta = \frac{4}{5}, \sin \beta = \frac{3}{5}$

$$\begin{aligned}
 \sin(\alpha - \beta) &= \sin \alpha \cos \beta - \cos \alpha \sin \beta \\
 &= \left(\frac{15}{17} \times \frac{4}{5} \right) - \left(\frac{8}{17} \times \frac{3}{5} \right) \\
 &= \frac{12}{17} - \frac{24}{85} = \frac{60 - 24}{85} \\
 \sin(\alpha - \beta) &= \frac{36}{85} \quad (A)
 \end{aligned}$$

$$\begin{aligned}
 \tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \\
 &= \left(\frac{15}{8} + \frac{3}{4} \right) \div \left[1 - \left(\frac{15}{8} \times \frac{3}{4} \right) \right]
 \end{aligned}$$

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$$= \left(\frac{15 + 6}{8} \right) \div \left(1 - \frac{45}{32} \right)$$

$$= \frac{21}{8} \div \frac{-13}{32}$$

$$= \frac{21}{8} \times \frac{32}{-13} = \frac{-84}{13}$$

$$\therefore \tan(x + \theta) = \frac{-84}{13} \quad (A)$$

$$\frac{11 - 3x}{x^2 + 2x - 3} = \frac{11 - 3x}{(x-1)(x+3)}$$

$$\therefore \frac{11 - 3x}{(x-1)(x+3)} = \frac{A}{x-1} + \frac{B}{x+3}$$

$$11 - 3x = A(x+3) + B(x-1)$$

$$\text{When } x = -3$$

$$11 - 3(-3) = A(-3+3) + B(-3-1)$$

$$20 = -4B$$

$$B = \frac{20}{-4} = -5, \therefore B = -5$$

$$\text{When } x = 1$$

$$11 - 3(1) = A(1+3) + B(1-1)$$

$$8 = 4A$$

$$A = \frac{8}{4} = 2$$

$$\therefore \frac{11 - 3x}{x^2 + 2x - 3} = \frac{2}{x-1} + \frac{-5}{x+3} \quad (B)$$

$$\frac{2x + 3}{(x-2)^2} = \frac{A}{x-2} + \frac{B}{(x-2)^2}$$

$$2x + 3 = A(x-2) + B = Ax - 2A + B$$

$$\text{When } x = 2$$

$$2(2) + 3 = A(2-2) + B$$

$$4 + 3 = B$$

$$B = 7$$

By comparing coefficients,
eqn = $A = 2$

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$$\frac{x^2-9}{(x-3)(x+3)} = \frac{x^2-3^2}{(x+3)(x-3)}$$

$$\frac{12}{(x-3)(x+3)} = \frac{A}{x-3} + \frac{B}{x+3}$$

$$12 = A(x+3) + B(x-3)$$

When $x = 3$

$$12 = A(3+3) + B(3-3)$$

$$12 = 6A$$

$$A = \frac{12}{6} = 2$$

When $x = -3$

$$12 = A(-3+3) + B(-3-3)$$

$$12 = -6B$$

$$B = \frac{12}{-6} = -2$$

$$\therefore \frac{12}{x^2-9} = \frac{2}{x-3} - \frac{2}{x+3} \quad (I)$$

NO QUESTION

$n =$

As we know; 0 1 2 3 4 5 6 7 8

$T_1 T_2 T_3 T_4 T_5 T_6 T_7 T_8$

$$36. (2a-5b)^8$$

The middle term is the fifth (5th) term.

$$\therefore T = 5$$

$$r+1 = T$$

$$r = T-1 = 5-1 = 4$$

$$T_{r+1} = {}^nC_r \cdot a^{n-r} \cdot b^r$$

$$T_5 = {}^8C_4 \cdot (2a)^{8-4} \cdot (-5b)^4$$

$$= {}^8C_4 (2a)^4 (-5b)^4$$

$$= 700,000 a^4 b^4 \quad (A)$$

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$$37. (2x-1)^8$$

$$\begin{aligned} T_{r+1} &= {}^nC_r \cdot a^{n-r} \cdot b^r \\ &= {}^8C_r \cdot (2x)^{8-r} \cdot (-1)^r \\ &= {}^8C_r \cdot 2^{8-r} \cdot (-1)^r \cdot x^{8-r} \end{aligned}$$

$$x^{8-r} = x^5$$

$$8-r = 5$$

$$r = 8-5 = 3$$

$$\therefore \text{8th } {}^8C_r \cdot 2^{8-r} \cdot (-1)^r,$$

By apply. To get the coefficient of x^5 ,

$$= {}^8C_3 \cdot 2^{8-3} \cdot (-1)^3$$

$$= {}^8C_3 \cdot 2^5 \cdot (-1)^3$$

$$= -1792 \quad (A)$$

$$38. (2x-x^2)^{12}$$

$$T_{r+1} = {}^nC_r \cdot a^{n-r} \cdot b^r$$

$$n = 12$$

$$= {}^{12}C_r \cdot a^{12-r} \cdot b^r$$

$$= {}^{12}C_r \cdot (2x)^{12-r} \cdot (-x^2)^r$$

$$= {}^{12}C_r \cdot (2)^{12-r} \cdot (-1)^r \cdot (x)^{12-r} \cdot (x^2)^r$$

$$x^{12-r+2r} = x^{20}$$

$$x^{12+r} = x^{20}$$

Comparing powers,

$$12+r = 20$$

$$r = 20-12 = 8$$

$$\therefore {}^{12}C_8 \times 2^{12-8} \cdot (-1)^8 = {}^{12}C_8 \cdot 2^4 \cdot (-1)^8$$

$$= 7920 \quad (C)$$

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$$\begin{aligned}
 (1+2x)^5 \\
 T_{r+1} &= {}^nC_r \cdot a^{n-r} \cdot b^r \\
 &= {}^5C_r \cdot (1)^{5-r} \cdot 2^r \cdot x^r \\
 x^r &= x^4 \\
 \therefore r &= 4 \\
 K &= {}^5C_4 (1)^{5-4} \cdot 2^4 \\
 &= {}^5C_4 \cdot (1) \cdot 2^4 \\
 K &= 80
 \end{aligned}$$

(D)

$$\sin \theta = \frac{3}{5} = \frac{\text{opp}}{\text{hyp}}$$



Applying Pythagoras Theorem.

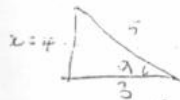
$$5^2 = 3^2 + x^2$$

$$25 - 9 = x^2$$

$$x = \sqrt{16} = 4$$

$$\therefore \cos \theta = \frac{4}{5} \quad \tan \theta = \frac{3}{4}$$

$$\cos \alpha = \frac{3}{5} = \frac{\text{Adj}}{\text{hyp}}$$



$$5^2 = 3^2 + x^2$$

$$25 - 9 = x^2$$

$$x = \sqrt{16} = 4$$

$$\sin \alpha = \frac{4}{5}, \quad \tan \alpha = \frac{4}{3}$$

$$\sin(\theta + \alpha) = \sin \theta \cos \alpha + \cos \theta \sin \alpha$$

$$= \left(\frac{3}{5} \cdot \frac{3}{5}\right) + \left(\frac{4}{5} \cdot \frac{4}{5}\right)$$

$$= \frac{9}{25} + \frac{16}{25}$$

$$= \frac{25}{25}$$

$$= 1$$

(A)

12

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$$\begin{aligned}
 41. \quad & \frac{\cos 60^\circ + \sin 30^\circ}{\sin 150^\circ} \\
 & \cos 60^\circ = \frac{1}{2} \quad \sin 30^\circ = \frac{1}{2} \\
 & \sin 150^\circ = \frac{1}{2} \\
 & \therefore = \frac{\frac{1}{2} + \frac{1}{2}}{\frac{1}{2}} = \frac{\frac{2}{2}}{\frac{1}{2}} \\
 & = \frac{2}{2} \times \frac{2}{1} \\
 & = 2 \quad (A)
 \end{aligned}$$

$$\begin{aligned}
 42. \quad & a \cos^2 \theta + b \sin^2 \theta = c \\
 & \text{Recall: } \sin^2 \theta + \cos^2 \theta = 1 \\
 & \sin^2 \theta = 1 - \cos^2 \theta \\
 & \cos^2 \theta = 1 - \sin^2 \theta \\
 & a(1 - \sin^2 \theta) + b \sin^2 \theta = c \\
 & a - a \sin^2 \theta + b \sin^2 \theta = c \\
 & a - c = a \sin^2 \theta - b \sin^2 \theta \\
 & a - c = \sin^2 \theta (a - b) \\
 & \sin^2 \theta = \frac{a - c}{a - b}
 \end{aligned}$$

From the initial equation:

$$a \cos^2 \theta + b \sin^2 \theta = c$$

$$a \cos^2 \theta + b(1 - \cos^2 \theta) = c$$

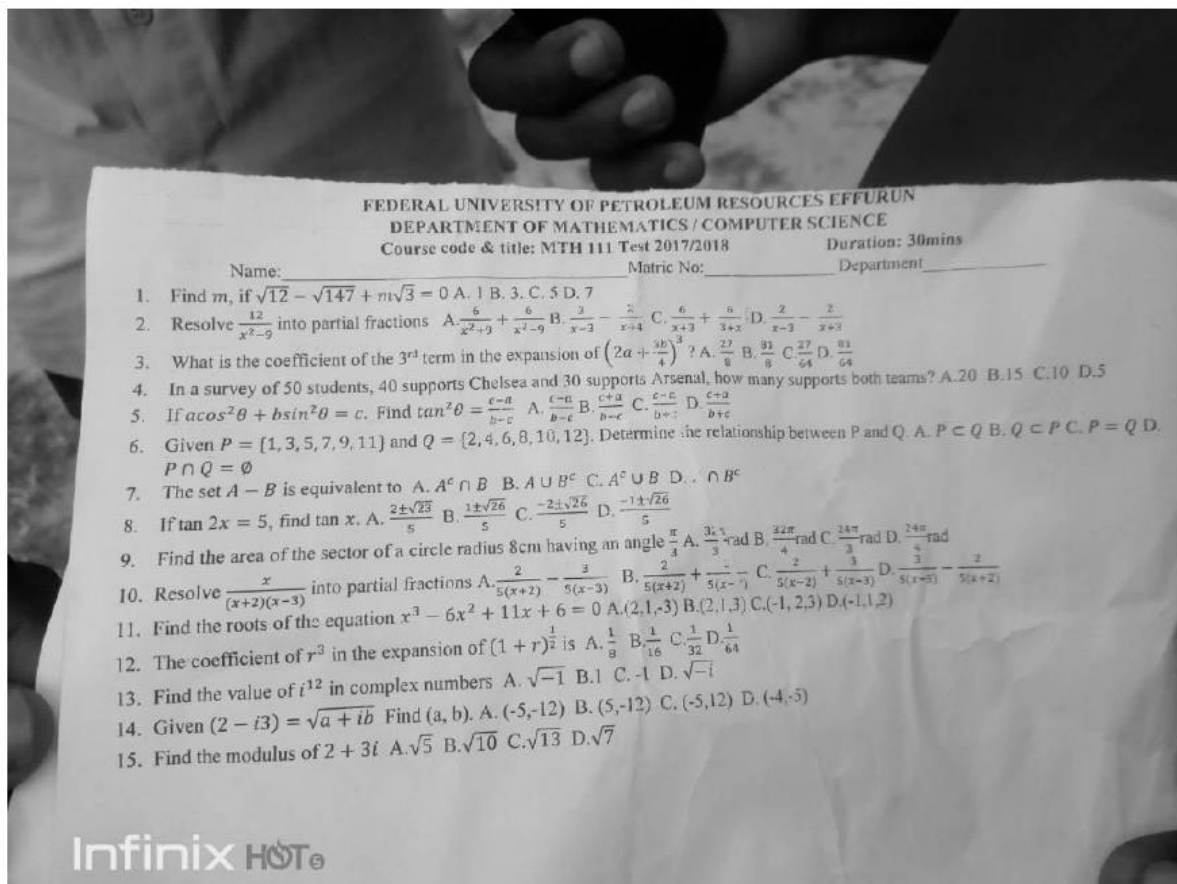
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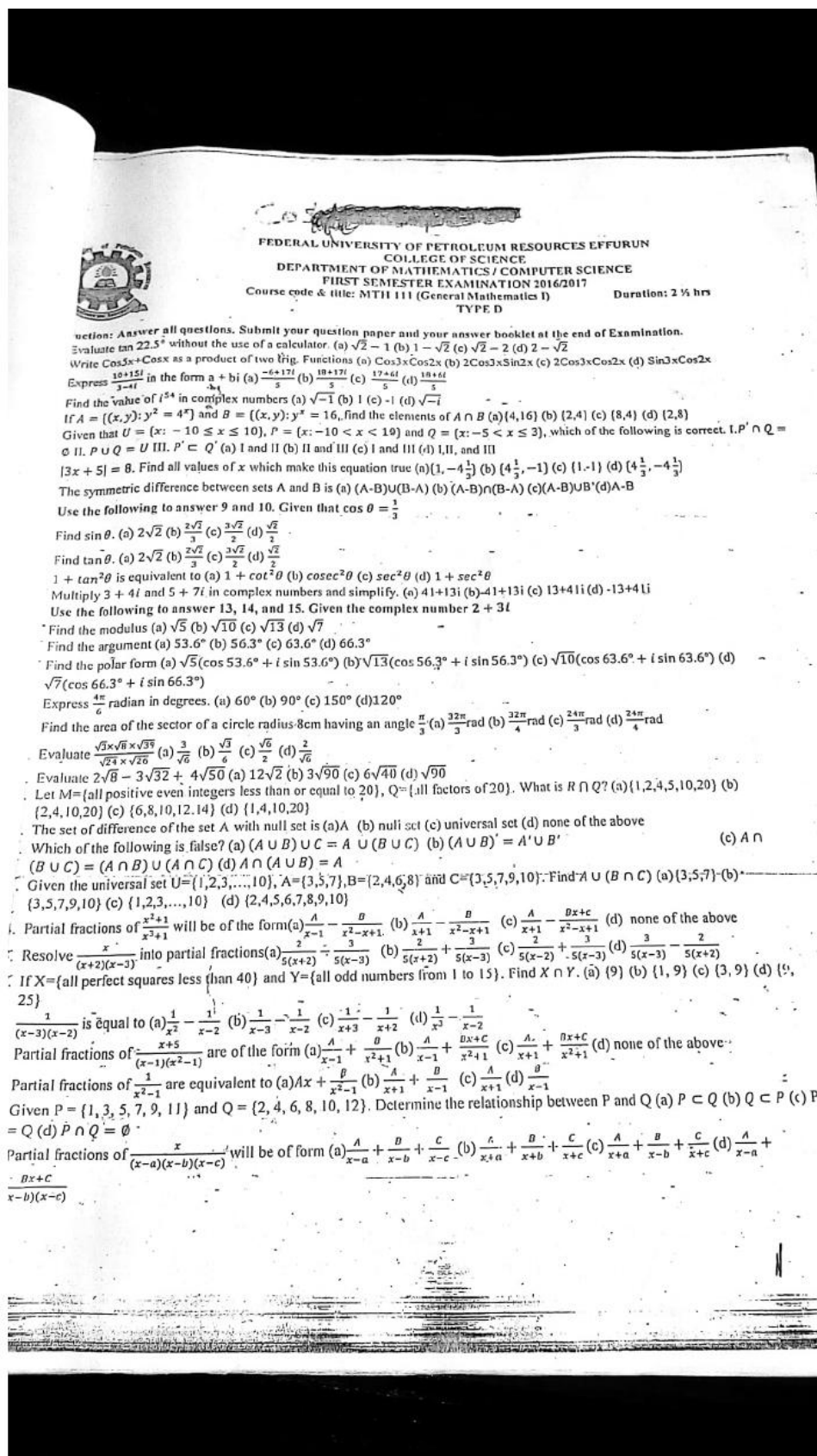


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32. Absolute value of the following integers -5, -1, 0, 2, 4 in descending order can be best expressed as (a) 5, 4, 2, 1, 0 (b) 0, 1, 2, 4, 5 (c) 2, 4, 0, -5, -1 (d) 0, -1, 2, -5, 4
33. The distance between -55 and 10 on a real number line is (a) 15 (b) -45 (c) -550 (d) 65
34. Let a, b , and c be real numbers. Which of the following is false (a) $a + (b + c) = (a + b) + c$ (b) $a(b + c) = ab + ac$ (c) $a(b - c) = ab - ac$ (d) $a(bc) = (ab)c$
35. Which of the following is not a subset of the set of real numbers (a) Natural numbers (b) Whole numbers (c) Irrational numbers (d) complex numbers
36. Which of the following numbers is farthest from the number 1 in the number line (a) -11 (b) -5 (c) 5 (d) 11
37. Express 35° in radian measure (a) $\frac{36}{7\pi}$ rad (b) $\frac{35}{5\pi}$ rad (c) $\frac{7\pi}{36}$ rad (d) $\frac{5\pi}{35}$ rad
38. Find the area of a triangle with sides 10cm, 7cm and 5cm. (a) 17.25cm^2 (b) 16.25cm^2 (c) 18.25cm^2 (d) 15.25cm^2
39. Evaluate $\tan 75^\circ$ without the use of a calculator. (a) $2 + \sqrt{3}$ (b) $2 + \sqrt{2}$ (c) $3 + \sqrt{2}$ (d) $3 + \sqrt{3}$
40. If $\tan 2\theta = 5$, find $\tan \theta$. (a) $\frac{2 \pm \sqrt{33}}{5}$ (b) $\frac{1 \pm \sqrt{26}}{5}$ (c) $\frac{-2 \pm \sqrt{26}}{5}$ (d) $\frac{-1 \pm \sqrt{26}}{5}$
41. The quadratic equation whose roots are $2 - \sqrt{2}$ and $2 + \sqrt{2}$ is (a) $x^2 + x - 2 = 0$ (b) $5x^2 - 3x + 11 = 0$ (c) $2x^2 - 3x + 7 = 0$ (d) $x^2 - 4x + 2 = 0$
42. Solve the equations: $m^2 - n^2 = 9$, $m - n = 3$. (a) (1, 2) (b) (3, 0) (c) (-1, 3) (d) (2, 5)
43. Solve for y in the equation, $\sqrt{y + 5} = 5 - \sqrt{y}$. (a) 1 (b) 2 (c) 3 (d) 4
44. For what value of k does the equation $3x^2 - 6x + k = 0$ have equal roots (a) 1 (b) 2 (c) 3 (d) 4
45. Find the value(s) of the number n such that twice its square is equal to three times the number plus 5. (a) $(-1, \frac{5}{2})$ (b) $(-2, \frac{1}{2})$ (c) $(-1, 2)$ (d) $(-2, \frac{3}{2})$
46. Given $ay^2 + by + c = 0$ if $b^2 - 4ac < 0$ then the roots are (a) distinct (b) equal (c) complex (d) undefined
47. If α and β are the roots of the equation $3x^2 + 5x - 2 = 0$. Find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$. (a) $\frac{5}{2}$ (b) $\frac{3}{2}$ (c) $\frac{1}{2}$ (d) $\frac{7}{2}$
48. What is the coefficient of the 3rd term in the expansion of $(2x + \frac{3x}{4})^3$? (a) $\frac{27}{8}$ (b) $\frac{81}{64}$ (c) $\frac{27}{64}$ (d) $\frac{81}{64}$
49. Find the coefficient of y^4 in the binomial expansion of $(1 - 2y)^6$. (a) 280 (b) 240 (c) 200 (d) 180
50. Find the value of n if the ratio of the coefficient of y^4 to that of y^3 in the binomial expansion of $(1 + 2y)^n$ is 3:1. (a) 3 (b) 6 (c) 9 (d) 12
51. Using Pascal triangle evaluate $(1 + \sqrt{3})^4 - (1 - \sqrt{3})^4$. (a) $9\sqrt{3}$ (b) $27\sqrt{3}$ (c) $32\sqrt{3}$ (d) $16\sqrt{3}$
52. The n th term of the series $\sum_{k=1}^n k^3$ is (a) $\frac{n(n+1)(2n+1)}{6}$ (b) $\frac{n(n^2-1)}{3}$ (c) $\frac{n(n+1)^2}{2}$ (d) $\frac{(2n-1)^2}{4}$
53. The value of n when $5^{2n} + 3n - 1$ is an integer multiple is (a) 2 (b) 3 (c) 5 (d) 7
54. The coefficient of x^3 in the expansion of $(1 + x)^{\frac{1}{2}}$ is (a) $\frac{1}{6}$ (b) $\frac{1}{16}$ (c) $\frac{1}{32}$ (d)
55. If $S_n = 2n(n + 3)$ is the sum of the first n terms of a series, find the 4th term. (a) 10 (b) 15 (c) 25 (d) 20
56. If set $A = \{2, 9, 5, 2\}$, then the number of subsets which can be formed from A is (a) 3 (b) 4 (c) 8 (d) 16
57. The cardinality of $B = \{5, 6, 3, 2, 3, 2\}$ is (a) 6 (b) 5 (c) 4 (d) 3
58. If Z represents the set of integers, then $\{y: -3 < y < 2\}$ is (a) $\{-3, 2\}$ (b) $\{0, 1\}$ (c) $\{-3, -2, -1, 0, 1, 2\}$ (d) $\{-2, -1, 0, 1\}$
59. In a survey of 50 newspapers, 40 read Champion and 30 read Guardian, how many read both papers? (a) 20 (b) 15 (c) 10 (d) 5
60. If $X = \{n^2 + 1: n = 0, 2, 3\}$ and $Y = \{n + 1: n = 2, 3, 5\}$ Find $X \cap Y$. (a) $\{1, 3\}$ (b) $\{5, 10\}$ (c) $\emptyset$ (d) $\{4, 6\}$
61. Given the sequence $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots$. Find its n th term. (a) $\frac{1}{2}(4n - 5)$ (b) $2n - 1$ (c) $5n - 2$ (d) $\frac{1}{2}(2n - 5)$
62. The third term of an arithmetic series is -5 and the fifth term is 4.5. Find the 13th term. (a) 33 (b) 42.5 (c) 14.5 (d) 47
63. If 19, x , y , -8 is an AP. What is (x, y) ? (a) (12, 7) (b) (7, 12) (c) (10, 1) (d) (15, 21)
64. The sum of the first twenty terms of the AP: $1\frac{1}{2}, 1\frac{1}{4}, 1, \dots$ is (a) -20 (b) -11 (c) -24.5 (d) -17.5
65. The sum of the first n terms of an AP is 252. If the first term is -16 and the last term is 72, find the number of terms n . (a) 9 (b) 10 (c) 6 (d) 7
66. Given an exponential sequence having 81 and 1 as the n th and 5 th terms respectively. Find the 7th term of the sequence. (a) $\frac{2}{3}$ (b) $\frac{1}{27}$ (c) $\frac{2}{9}$ (d) $\frac{1}{27}$
67. Evaluate the sum to infinity of the series: $1 + t + t^2 + t^3 + \dots$ (a) t^∞ (b) $\frac{2}{1-t}$ (c) $\frac{1}{1-t}$ (d) undefined
68. The n th term of the sequence 2, 6, 12, 20, 30, ... is (a) $n^2 - n$ (b) $\frac{1}{2}n + 1$ (c) $2n + \frac{n^2}{2}$ (d) $n(n + 1)$
69. A sequence of numbers $T_1, T_2, T_3, \dots, T_n$ satisfies the relation $T_{n+1} + n^2 = nT_n + 2$ for all integers $n \geq 1$. If $T_1 = 2$, Find $T = (T_2, T_3)$. (a) (1, 2) (b) (2, 3) (c) (3, 4) (d) (1, 3)
70. Find the roots of the equation: $x^3 - 2x^2 + x + 2 = 0$ (a) (1, 2, 1) (b) (-1, 2, 3) (c) (1, -2, 2) (d) (-1, -1, 2)

A. $\tan 22.5$

From $\tan \alpha = \frac{2 \tan \frac{\alpha}{2}}{1 - \tan^2 \frac{\alpha}{2}}$

Let $\alpha = 45^\circ$.

$$\tan 45 = \frac{2 \tan \left(\frac{45}{2}\right)}{1 - \tan^2 \left(\frac{45}{2}\right)} \Rightarrow \frac{2 \tan 22.5}{1 - \tan^2 22.5}$$

Let $\tan 22.5 = t$.

$$\tan 45 = \frac{2t}{1 - t^2} \quad \text{from Special angles}$$

$\tan 45 = 1$.

$$1 = \frac{2t}{1 - t^2}$$

$$\therefore t^2 + 2t - 1 = 0$$

Using quadratic formula,

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$t = \frac{-2 \pm \sqrt{4 - 4(1)(-1)}}{2 \cdot 1}$$

$$t = \frac{-2 \pm \sqrt{4 + 4}}{2}$$

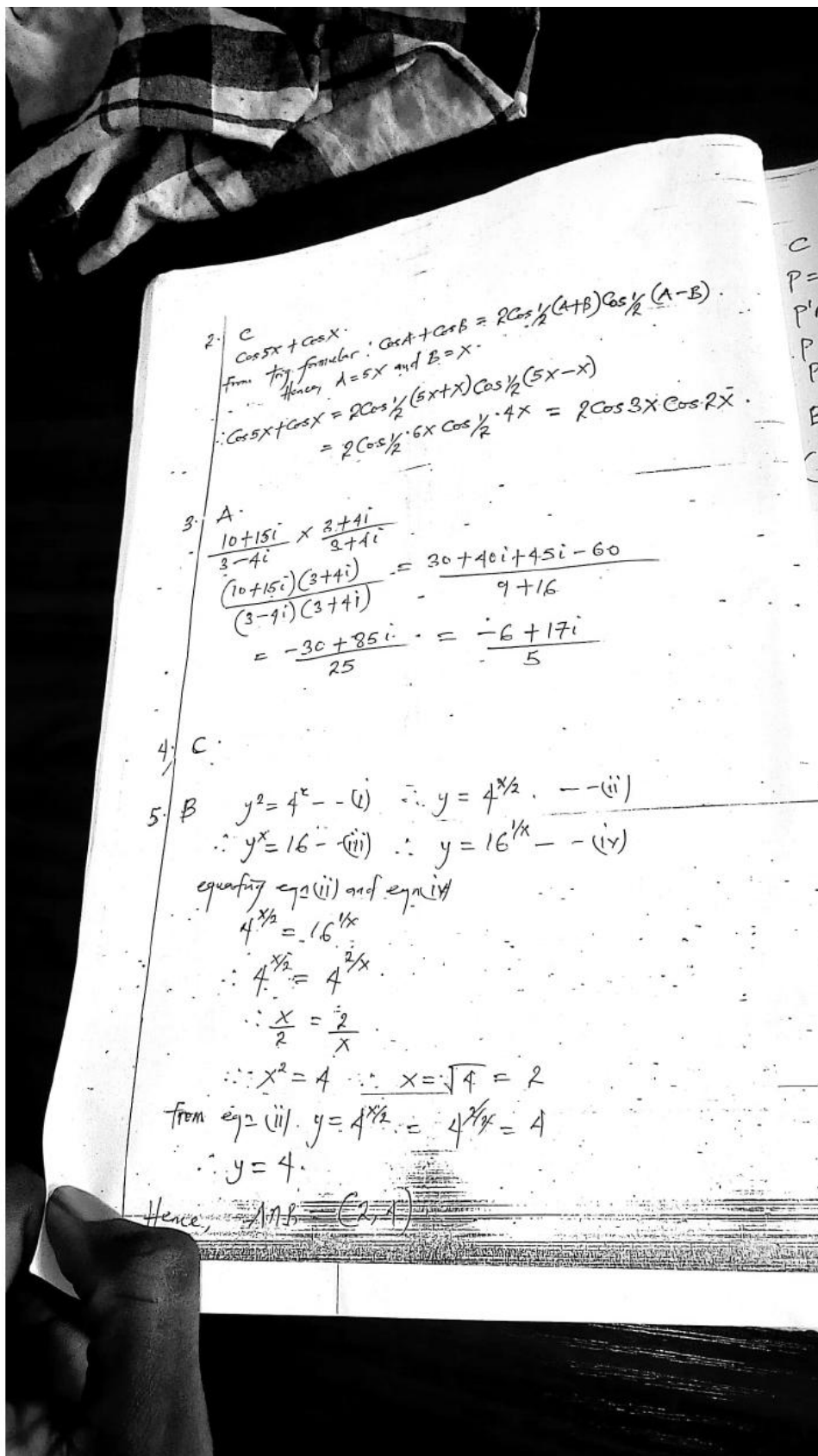
$$t = \frac{-2 \pm \sqrt{8}}{2}$$

$$t = \frac{-2 \pm 2\sqrt{2}}{2}, \therefore -1 \pm \sqrt{2}$$

Hence, $t = -1 + \sqrt{2}$ or $-1 - \sqrt{2}$

$t = \sqrt{2} - 1$ i.e. A.

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C $U = \{-10, \dots, 10\}$

$P = \{-9, \dots, 9\}$, $Q = \{-4, \dots, 3\}$

$P \cap Q = \emptyset$ True.

$P \cup Q = U$ False

$P \subset Q$ True.

B $|3x+5| = 8$

$(|3x+5|)^2 = 8^2$

$9x^2 + 30x - 39 = 64$

$3x^2 + 10x - 13 = 0$

$(3x^2 + 3x) - (13x - 13) = 0$

$3x(x+1) - 13(x+1) = 0$

$(3x-13)(x+1) = 0$

$\therefore x = -1$ or $3x = 13$

$(4\frac{1}{3}, -1)$ $\therefore x = 4\frac{1}{3}$

D

B

Recall $\cos \theta = \frac{\text{Adj}}{\text{Hyp}} = \frac{1}{3}$

$\therefore \text{opp} = \sqrt{\text{Hyp}^2 - \text{Adj}^2}$
 $= \sqrt{3^2 - 1^2} = \sqrt{9-1}$
 $\therefore \text{opp} = \sqrt{8} = 2\sqrt{2}$

$\therefore \sin \theta = \frac{\text{opp}}{\text{Hyp}} = \frac{2\sqrt{2}}{3}$

A

$\tan \theta = \frac{\text{opp}}{\text{Adj}} = \frac{2\sqrt{2}}{1} = 2\sqrt{2}$

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11. C

from $\sin^2 t + \cos^2 t = 1$
dividing through by $\cos^2 t$

$$\frac{\sin^2 t}{\cos^2 t} + \frac{\cos^2 t}{\cos^2 t} = \frac{1}{\cos^2 t}$$

$$\therefore \tan^2 t + 1 = \sec^2 t$$

12. D

$$(3+4i) \cdot (5+7i)$$

$$15+21i+20i-28 = -13+41i$$

13. C

$$|2+3i| = \sqrt{2^2+3^2} = \sqrt{4+9} = \sqrt{13}$$

14. B

Argument = θ

$$\therefore \tan \theta = y/x = 3/2$$

$$\therefore \theta = \tan^{-1}(3/2) = 56.3^\circ$$

15. B

Polar form = $r(\cos t + i \sin t)$

$$\therefore r = |2+3i| = \sqrt{13}$$

$$\therefore \text{Here} = \sqrt{13} (\cos 56.3^\circ + i \sin 56.3^\circ)$$

16. D

$$\therefore \text{degree} = X_{\text{rad}} \cdot \frac{180}{\pi}$$

$$= \frac{4\pi}{6} \times \frac{180}{\pi} = 120^\circ$$

17. A

$$A_{\text{sector}} = \frac{\theta}{360} \times \pi r^2$$

$$\theta = \pi/3, \quad r = 8 \text{ cm}$$

$$\text{But, } \pi = 180^\circ \therefore 2\pi = 360^\circ$$

$$\therefore \frac{\pi}{3 \times 2\pi} \times \pi \cdot 8^2 = \frac{\pi}{6} \times 64 = \frac{32\pi}{3} \text{ rad}$$

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A.

$$\frac{\sqrt{3} \times \sqrt{9} \times \sqrt{54}}{\sqrt{24} \times \sqrt{26}} = \frac{\sqrt{3} \times 2\sqrt{2} \times \sqrt{3} \times \sqrt{3}}{2\sqrt{6} \times \sqrt{3} \times \sqrt{2}} = \frac{3}{\sqrt{6}}$$

A.

$$\begin{aligned} & 2\sqrt{8} - 3\sqrt{32} + 4\sqrt{50} \\ &= 2\sqrt{4 \times 2} - 3\sqrt{16 \times 2} + 4\sqrt{25 \times 2} \\ &= 2\sqrt{4} \cdot \sqrt{2} - 3\sqrt{16} \cdot \sqrt{2} + 4\sqrt{25} \cdot \sqrt{2} \\ &= 4\sqrt{2} - 12\sqrt{2} + 20\sqrt{2} \\ &= 12\sqrt{2} \end{aligned}$$

A.

A.

$$A - \emptyset = A$$

B.

$$(A \cup B)' = A' \cap B'$$

A

$$B \cap C = \{\emptyset\}$$

$$\therefore A \cup (B \cap C) = A \cup \emptyset = A$$

$$\therefore \text{And } A = \{3, 5, 7\}$$

C

$$\frac{x^2+1}{x^3+1} = \frac{x^2+1}{(x+1)(x^2-x+1)} = \frac{A}{x+1} + \frac{Bx+C}{(x^2-x+1)} \text{ in partial fraction}$$

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$$25. \frac{B}{x} = \frac{A}{(x+2)} + \frac{B}{(x-3)}$$

$$\frac{A(x-3) + B(x+2)}{(x+2)(x-3)}$$

To find A, put $x = -2$

$$\therefore A(-2-3) = -2$$

$$\therefore -5A = -2 \therefore A = \frac{2}{5}$$

To find B, put $x = 3$

$$\therefore B(3+2) = 3 \therefore B = \frac{3}{5}$$

$$\text{Hence, } \frac{x}{(x+2)(x-3)} = \frac{2}{5(x+2)} + \frac{3}{5(x-3)}$$

$$26. B$$

$$X = \{1, 4, 9, 16, 25, 36\} \text{ and } Y = \{1, 3, 5, 7, 9, 11, 13, 15\}$$

$$\therefore X \cap Y = \{1, 9\}$$

$$27. B$$

$$\frac{1}{(x-3)(x-2)} = \frac{A}{(x-3)} + \frac{B}{(x-2)}$$

$$\frac{A(x-2) + B(x-3)}{(x-3)(x-2)}$$

To find A, put $x = 3$

$$A(3-2) = 1 \therefore A = 1$$

To find B, put $x = 2$

$$B(2-3) = 1 \therefore B = -1$$

$$\therefore \frac{1}{(x-3)(x-2)} = \frac{1}{x-3} - \frac{1}{x-2}$$

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$$D.$$

$$\frac{x+5}{(x-1)(x^2-1)} = \frac{x+5}{(x-1)(x-1)(x+1)}$$

$$= \frac{A}{(x-1)^2} + \frac{B}{(x-1)} + \frac{C}{(x+1)}$$

$$B.$$

$$\frac{1}{x^2-1} = \frac{1}{(x+1)(x-1)} = \frac{A}{(x+1)} + \frac{B}{(x-1)}$$

D.

A.

A

Absolute implies: $\{5, -1, 1, 2, 4\}$.

$= 5, 1, 0, 2, 4$

hence in descending Order: $5, 4, 2, 1, 0$.

B.

C

C

As natural numbers, whole numbers, integers and complex numbers are all a subset of rational numbers which is a set of real numbers.

A.

C

$$\text{From degree} = X_{\text{rad}} \cdot \frac{180}{\pi}$$

$$\therefore X_{\text{rad}} = \frac{\text{degree} \times \pi}{180} = \left(\frac{35^\circ \times \pi}{180} \right) \text{rad}$$

$$= \frac{7\pi}{36} \text{rad}$$

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32. B.

Using Heron's formula,

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{where } s = \frac{a+b+c}{2}$$

$$\therefore \text{given: } a = 10\text{cm}, b = 7\text{cm}, c = 5\text{cm}$$

$$s = \frac{10+7+5}{2} = 11\text{cm}$$

$$A = \sqrt{11(11-10)(11-7)(11-5)}$$

$$A = \sqrt{264} = 16.25\text{cm}^2$$

39. A.

$$\tan 75^\circ = \tan (45^\circ + 30^\circ)$$

$$\text{Recall, } \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\therefore \tan(45^\circ + 30^\circ) = \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ}$$

$$\text{Recall from special angles, } \tan 45^\circ = 1; \tan 30^\circ = \frac{\sqrt{3}}{3}$$

$$\therefore \tan 75^\circ = \frac{1 + \frac{\sqrt{3}}{3}}{1 - 1 \cdot \frac{\sqrt{3}}{3}} = \frac{3 + \sqrt{3}}{3 - \sqrt{3}} \times \frac{3 + \sqrt{3}}{3 + \sqrt{3}} \quad \left(\text{Hint: Using rationalization} \right)$$

$$\therefore \tan 75^\circ = \frac{12 + 6\sqrt{3}}{6} = 2 + \sqrt{3}$$

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D

$$\text{Recall. } \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$\therefore \text{let } \tan \theta = x$$

$$\text{given } \tan 2\theta = 5$$

$$\therefore 5 = \frac{2x}{1 - x^2}$$

$$\therefore 5 - 5x^2 = 2x^2$$

$$\therefore 5x^2 + 2x - 5 = 0$$

using quadratic formula,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\therefore x = \frac{-2 \pm \sqrt{4 - 4(5)(-5)}}{2(5)} = \frac{-2 \pm \sqrt{104}}{10}$$

$$= \frac{-2 \pm 2\sqrt{26}}{10} = \frac{-1 \pm \sqrt{26}}{5}$$

$$\text{Hence } x = \tan \theta = \frac{-1 \pm \sqrt{26}}{5}$$

D

Quadratic equation with known roots:

$$x^2 - x(\text{Sum of root}) + (\text{Product of root}) = 0$$

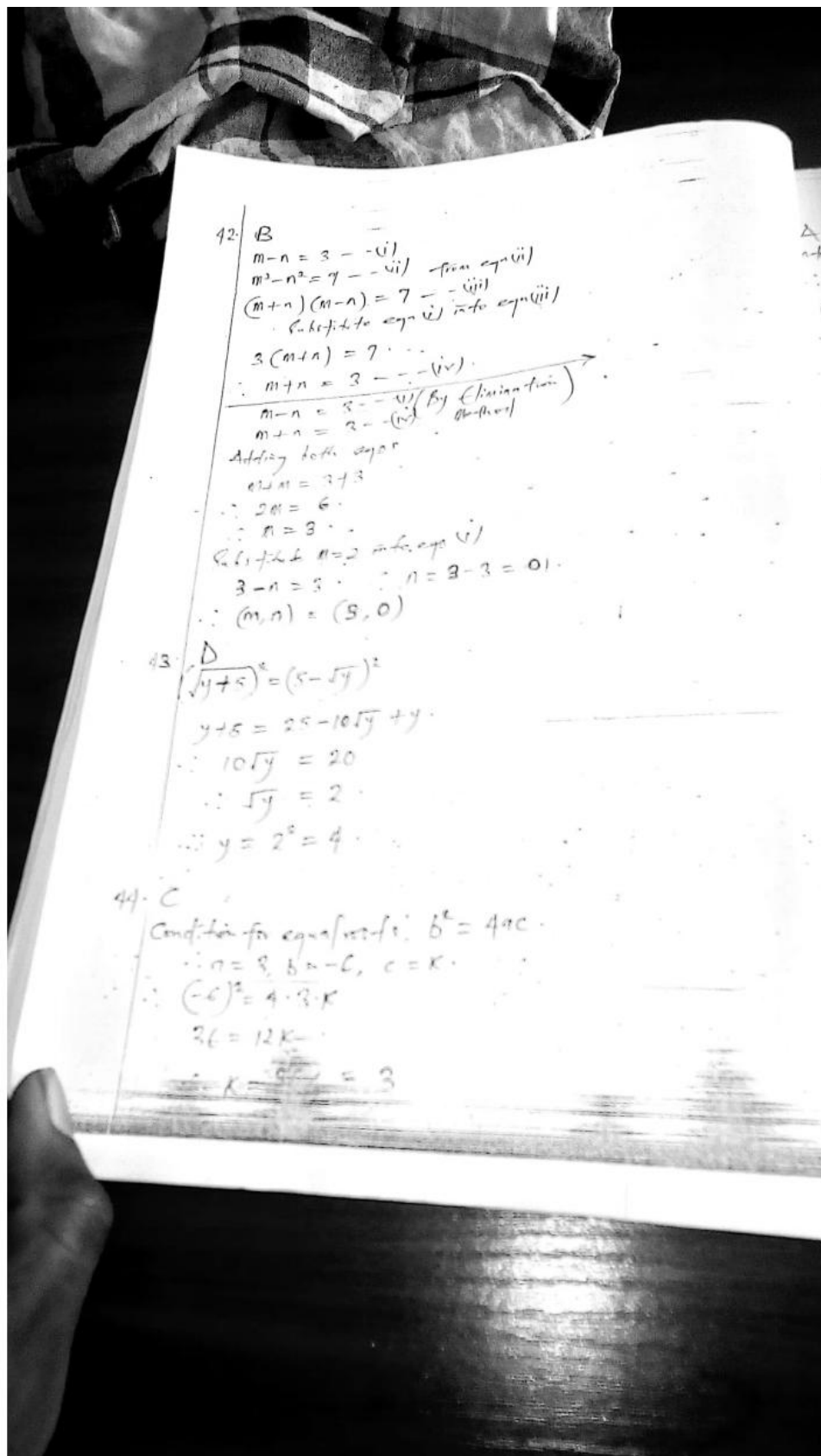
$$\therefore x^2 - x(\alpha + \beta) + \alpha \cdot \beta = 0$$

$$\alpha = 2 - \sqrt{2}, \quad \beta = 2 + \sqrt{2}$$

$$x^2 - x(2 - \sqrt{2} + 2 + \sqrt{2}) + (2 - \sqrt{2})(2 + \sqrt{2}) = 0$$

$$\therefore x^2 - 4x + 2 = 0$$

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A.

Interpretation: $2n^2 = 3n + 5$.

$$\therefore 2n^2 - 3n - 5 = 0$$

$$\therefore (2n^2 + 2n) - (5n - 5) = 0$$

$$\therefore 2n(n+1) - 5(n+1) = 0$$

$$\therefore (n+1)(2n-5) = 0$$

$$\therefore n = -1, \quad 2n-5 = 0$$

$$2n = 5 \quad \therefore n = \frac{5}{2} = 1\frac{1}{2}$$

$$= (-1, \frac{5}{2})$$

C

Discriminant $D = b^2 - 4ac$

$D \geq 0$, implies the roots are real.

$D = 0$, implies the roots are equal

$D > 0$, implies the roots are real and different.

$D < 0$, implies the roots are imaginary or complex.

A.

$a = 3, b = 5, c = -2$

recall, $\alpha + \beta = -\frac{b}{a} = -\frac{5}{3}$

$\alpha \cdot \beta = \frac{c}{a} = -\frac{2}{3}$

$$\therefore \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha \cdot \beta} = \frac{-\frac{5}{3}}{-\frac{2}{3}} = \frac{5}{2}$$

A:

qs $r = T - 1$

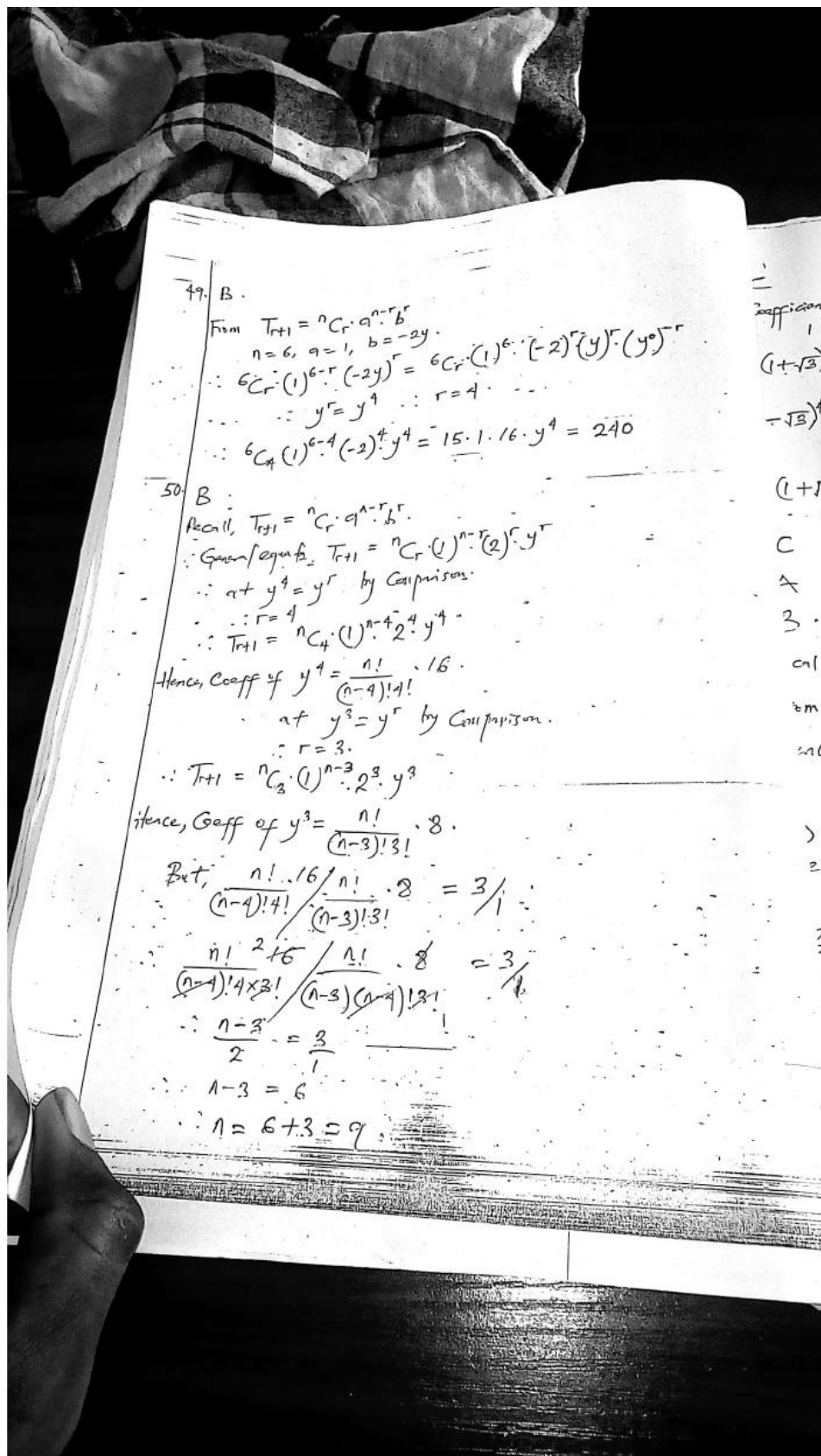
$T = 3, \therefore r = 3 - 1 = 2$

$T_{r+1} = {}^nC_r \cdot a^{n-r} b^r$. . qs $a = 2x, b = \frac{3y}{4}, n = 3$

$= {}^3C_2 (2x)^{3-2} \left(\frac{3y}{4}\right)^2$

$3 \cdot 2 \cdot x \cdot \frac{9}{16} y^2 = \frac{27}{8} xy^2$

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C

Coefficient of the fourth power using Pascal's triangle:

1 4 6 4 1

$$\begin{aligned} (1+\sqrt{3})^4 &= 1 + 4\sqrt{3} + 6(\sqrt{3})^2 + 4(\sqrt{3})^3 + (\sqrt{3})^4 \\ &= 1 + 4\sqrt{3} + 18 + 12\sqrt{3} + 9 = 28 + 16\sqrt{3} \end{aligned}$$

$$\begin{aligned} (1-\sqrt{3})^4 &= 1 + 4(-\sqrt{3}) + 6(-\sqrt{3})^2 + 4(-\sqrt{3})^3 + (-\sqrt{3})^4 \\ &= 1 - 4\sqrt{3} + 18 - 12\sqrt{3} + 9 = 28 - 16\sqrt{3} \end{aligned}$$

$$\begin{aligned} (1+\sqrt{3})^4 - (1-\sqrt{3})^4 &= (28 + 16\sqrt{3}) - (28 - 16\sqrt{3}) \\ &= 28 + 16\sqrt{3} - 28 + 16\sqrt{3} = 32\sqrt{3} \end{aligned}$$

C

A

B

$$\text{Recall, } (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{6}x^3 + \dots$$

$$\text{From } (1+x)^{1/2}, \therefore n = 1/2$$

$$\begin{aligned} \text{Hence Coeff of } x^3 &= \frac{n(n-1)(n-2)}{6} = \frac{1/2(-1/2)(-3/2)}{6} \\ &= 1/16 \end{aligned}$$

D

$$\text{Recall } T_n = S_n - S_{n-1}$$

$$\therefore T_4 = S_4 - S_3$$

$$\text{But } S_n = 2n(n+3)$$

$$\therefore S_4 = 2 \cdot 4 \cdot (4+3) = 56$$

$$S_3 = 2 \cdot 3 \cdot (3+3) = 36$$

$$\therefore T_4 = 56 - 36$$

$$= 20$$

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56. C
 From $A = \{2, 3, 5, 2\} \therefore n(A) = 3$
 $P = 2^A = 2^3 = 8$ - as 1 diff can be repeated element again.

57. A

58. D

59. A

$n(C) = 40$ $n(G) = 30$ $n(C \cap G) = 20$
 $\therefore 40 - x + x + 30 - x = 50$
 $70 - x = 50$
 $\therefore x = 70 - 50 = 20$
 $\therefore n(C \cap G) = 20$

60. C
 $X = \{1, 5, 10\}$, $Y = \{3, 4, 6\}$
 $\therefore X \cap Y = \emptyset$

61. A

$a = T_1 = -\frac{1}{2}$
 $d = T_2 - T_1 = T_3 - T_2 = \frac{3}{2} + \frac{1}{2} = 2$
 Recall, $T_n = a + (n-1)d$
 $\therefore T_n = -\frac{1}{2} + 2(n-1) = 2n - \frac{5}{2}$
 $\therefore T_n = \frac{1}{2}(4n-5)$

62. B

$T_3 = a + 2d = -5$ - (i)
 $T_5 = a + 4d = 4.5$ - (ii)
 Subtracting eqn (i) from eqn (ii)
 $Ad - 2d = 4.5 + 5$
 $\therefore 2d = 9.5$ - from eqn (i), $a + 2d = -5$
 $\therefore d = 4.75$ $\therefore a = -5 - 2(4.75)$
 $\therefore a = -5 - 9.5$
 $\therefore a = -14.5$
 $T_{13} = a + 12d = -14.5 + 12(4.75)$
 $= 57 - 14.5$
 $T_{13} = 42.5$

$a = 17$
 $= a + 3d =$
 Substitute
 $17 + 3d =$
 $3d = -2$
 $\therefore T_2 = a +$
 $= T_3 = a +$
 $(x, y) =$
 $= 1\frac{1}{2} =$
 $= T_5 =$
 $T_n = \frac{1}{2}$
 $S_{20} =$
 A
 $T_n = 2.5$
 Recall,

B
 equal
 T_5

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C

$$T_1 = a = 19 \text{ --- (i)}$$

$$T_4 = a + 3d = -8 \text{ --- (ii)}$$

Substitute eqn(i) into eqn(ii)

$$19 + 3d = -8$$

$$3d = -27 \therefore d = -9$$

$$x = T_2 = a + d = 19 - 9 = 10$$

$$y = T_3 = a + 2d = 19 + 18 = 1$$

$$\therefore (x, y) = (10, 1)$$

D

$$a = 1\frac{1}{2} = \frac{3}{2}$$

$$d = T_2 - T_1 = \frac{5}{4} - \frac{3}{2} = -\frac{1}{4}$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$\therefore S_{20} = \frac{20}{2} [2 \cdot \frac{3}{2} + (20-1) \cdot -\frac{1}{4}] = 30 - 47.5$$

$$\therefore S_{20} = -17.5$$

A

$$S_n = 252, a = -16, l = 72$$

Recall, $S_n = \frac{n}{2} [a + l]$

$$\frac{n}{2} [-16 + 72] = 252$$

$$\therefore n = 9$$

B

Recall for G.P, $T_n = ar^{n-1}$

$$\therefore T_1 = a = 81$$

$$T_5 = ar^4 = 1$$

$$\therefore T_5/T_1 = \frac{ar^4}{a} = \frac{1}{81} \therefore r = \frac{1}{3}$$

$$\therefore T_7 = ar^6 = 81 \left(\frac{1}{3}\right)^6 = \frac{1}{9}$$

C

$$a = 1, r = t$$

$$S_{\infty} = \frac{a}{1-r} = \frac{1}{1-t}$$

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68. D

$$a = 2, d = n + 2$$

$$\text{Recall, } T_n = a + (n-1)d$$

$$\therefore T_n = 2 + (n-1)(n+2)$$

$$\therefore T_n = 2 + n^2 + n - 2$$

$$\therefore T_n = n^2 + n$$

$$\text{Hence, } T_n = n(n+1)$$

69. C

$$\text{From: } T_{n+1} + n^2 = nT_n + 2$$

$$\text{at } n=1, T_1 = 2$$

$$\therefore T_{1+1} + 1^2 = 1 \cdot T_1 + 2$$

$$\therefore T_2 + 1 = 1 \cdot T_1 + 2$$

$$\therefore T_2 = T_1 + 2 - 1 = 2 + 2 - 1$$

$$\therefore T_2 = 4 - 1 = 3$$

$$\text{at } n=2$$

$$T_{2+1} + 2^2 = 2T_2 + 2$$

$$T_3 + 4 = 2T_2 + 2$$

$$\therefore T_3 = 2T_2 + 2 - 4$$

$$\therefore T_3 = 4$$

$$\therefore (T_2, T_3) = (3, 4)$$

70. D

From information, $x+1$ is a factor

$$x^3 - 2x^2 + x + 2$$

$$\text{Hence, } x^3 - x + 2$$

$$x+1 \overline{) x^3 - 2x^2 + x + 2}$$

$$\underline{x^3 + x^2}$$

$$\underline{-x^2 + x + 2}$$

$$\underline{-x^2 + x}$$

$$\underline{2x + 2}$$

$$\underline{2x + 2}$$

$$\therefore \text{factors: } (x+1)(x^2 - x + 2)$$

But $(x^2 - x + 2)$ cannot be factor

Hence, the problem assumes

$$x^2 - x - 2 = 0$$

$$(x^2 + x) - (2x - 2) = 0$$

$$x(x+1) - 2(x-1) = 0$$

$$(x+1)(x-2) = 0$$

$$\text{Hence, factors: } (x+1)(x-1)(x-2) = 0$$

$$\therefore (-1, -1, 2)$$

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